

Descida do gradiente

Francisco Rodrigues

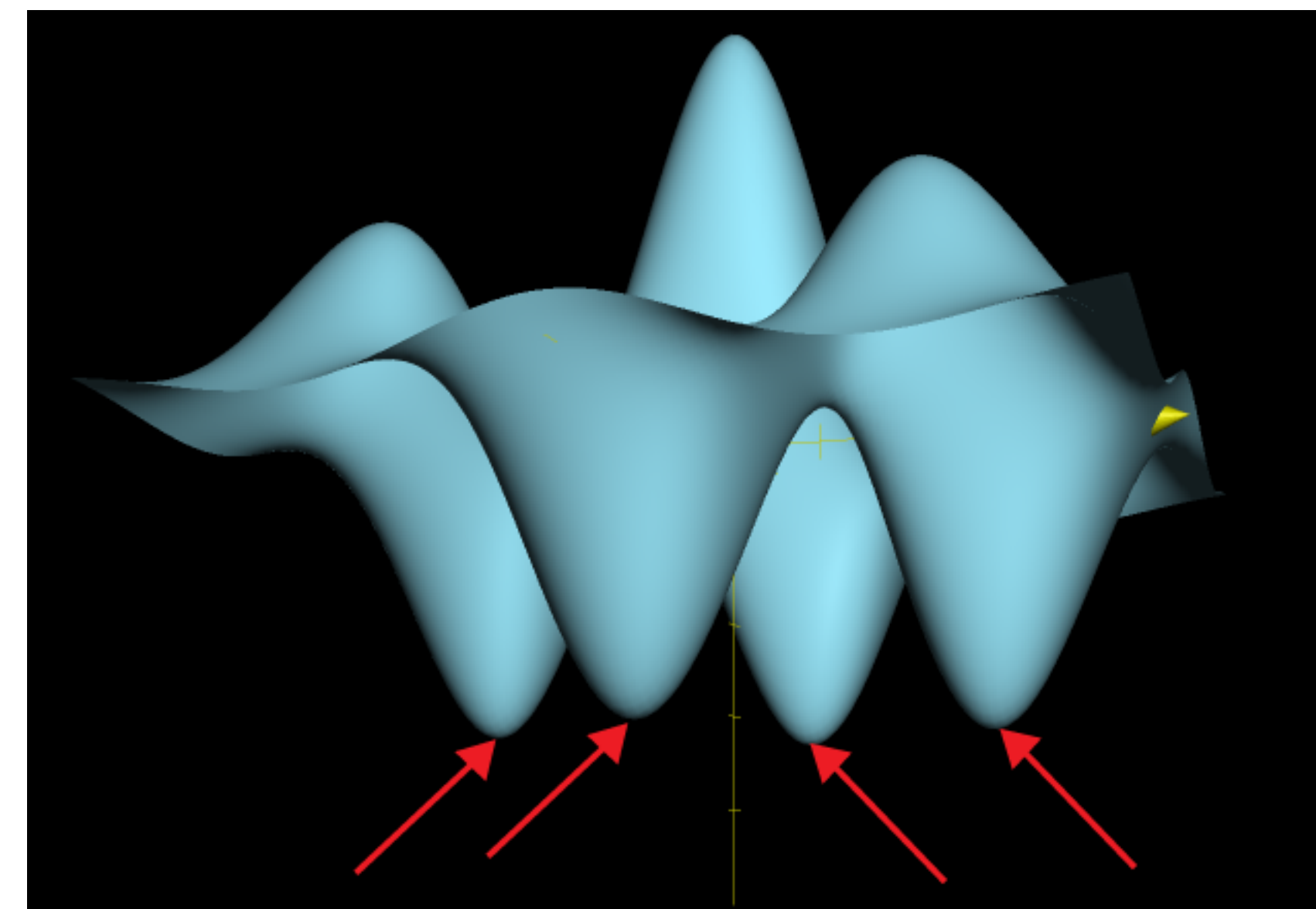
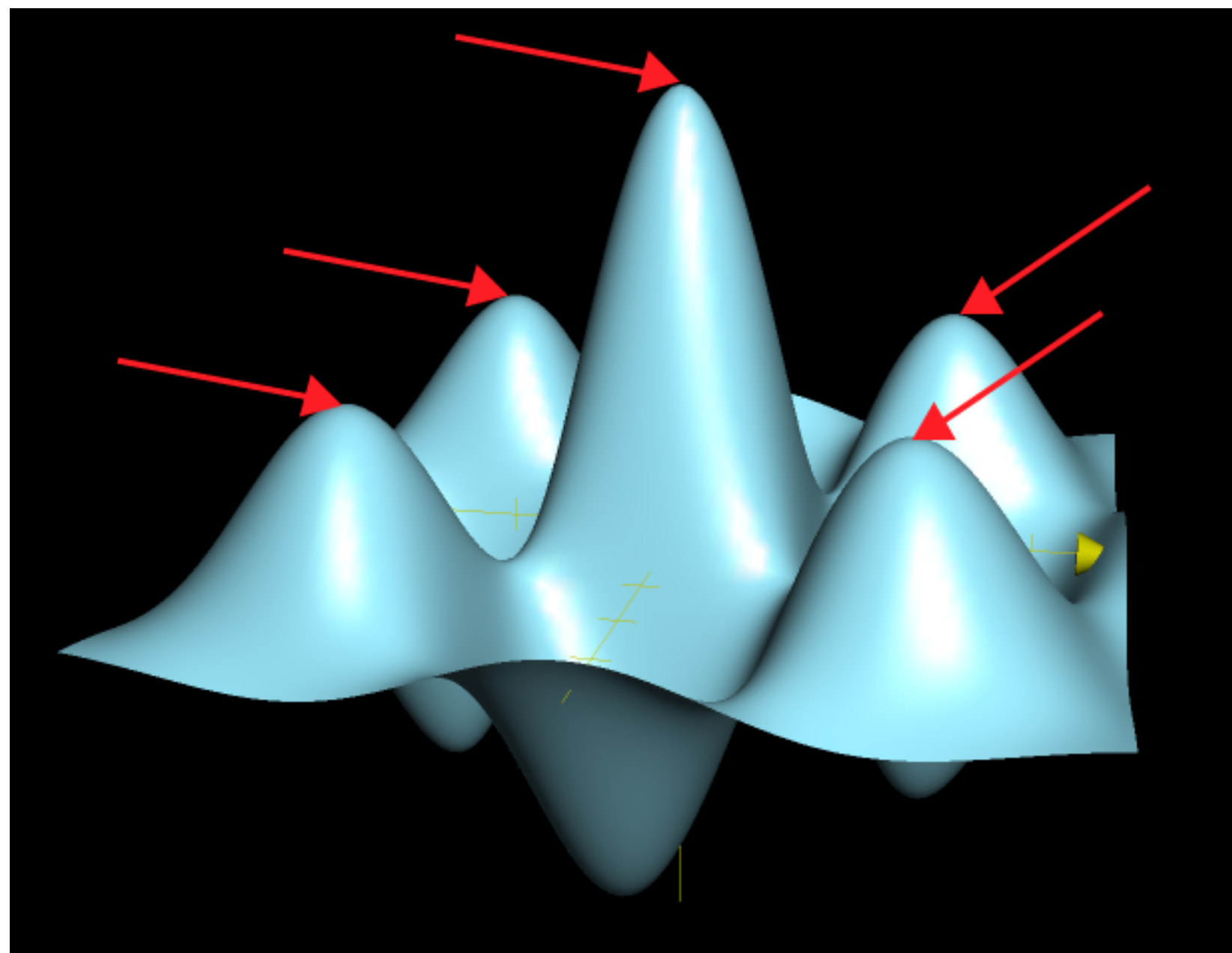
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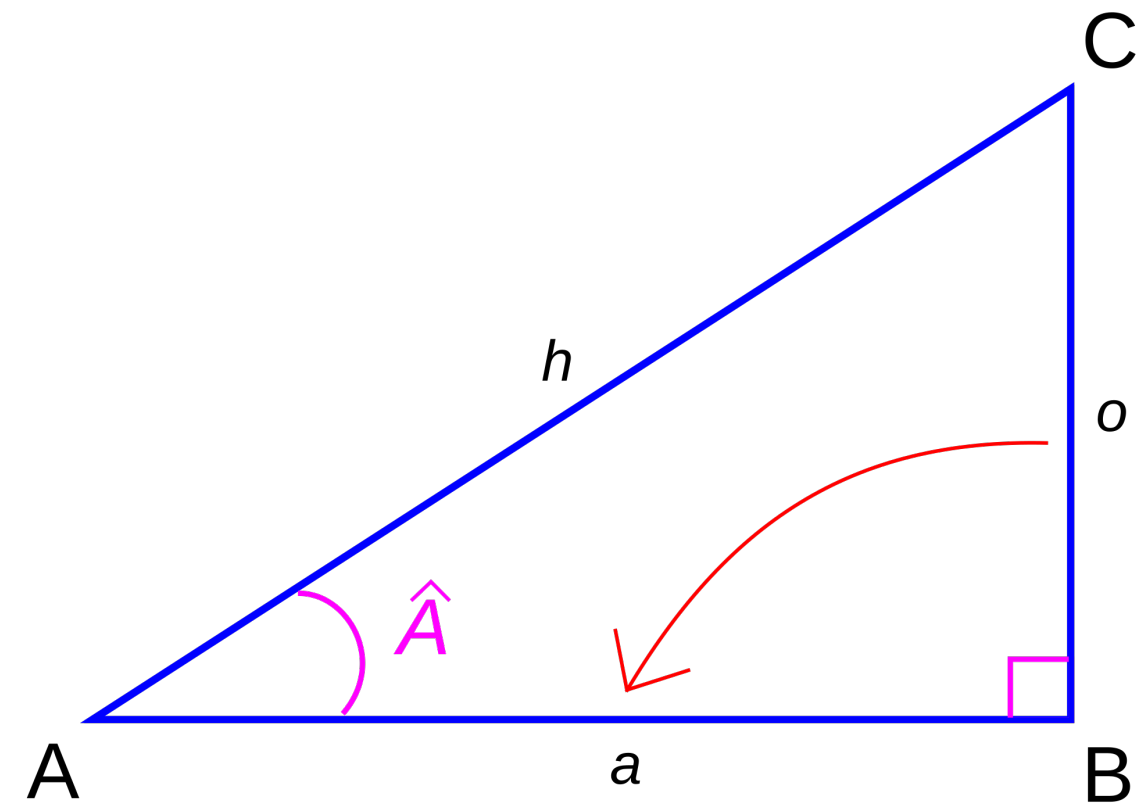
Máximo e mínimo de uma função

Máximos locais

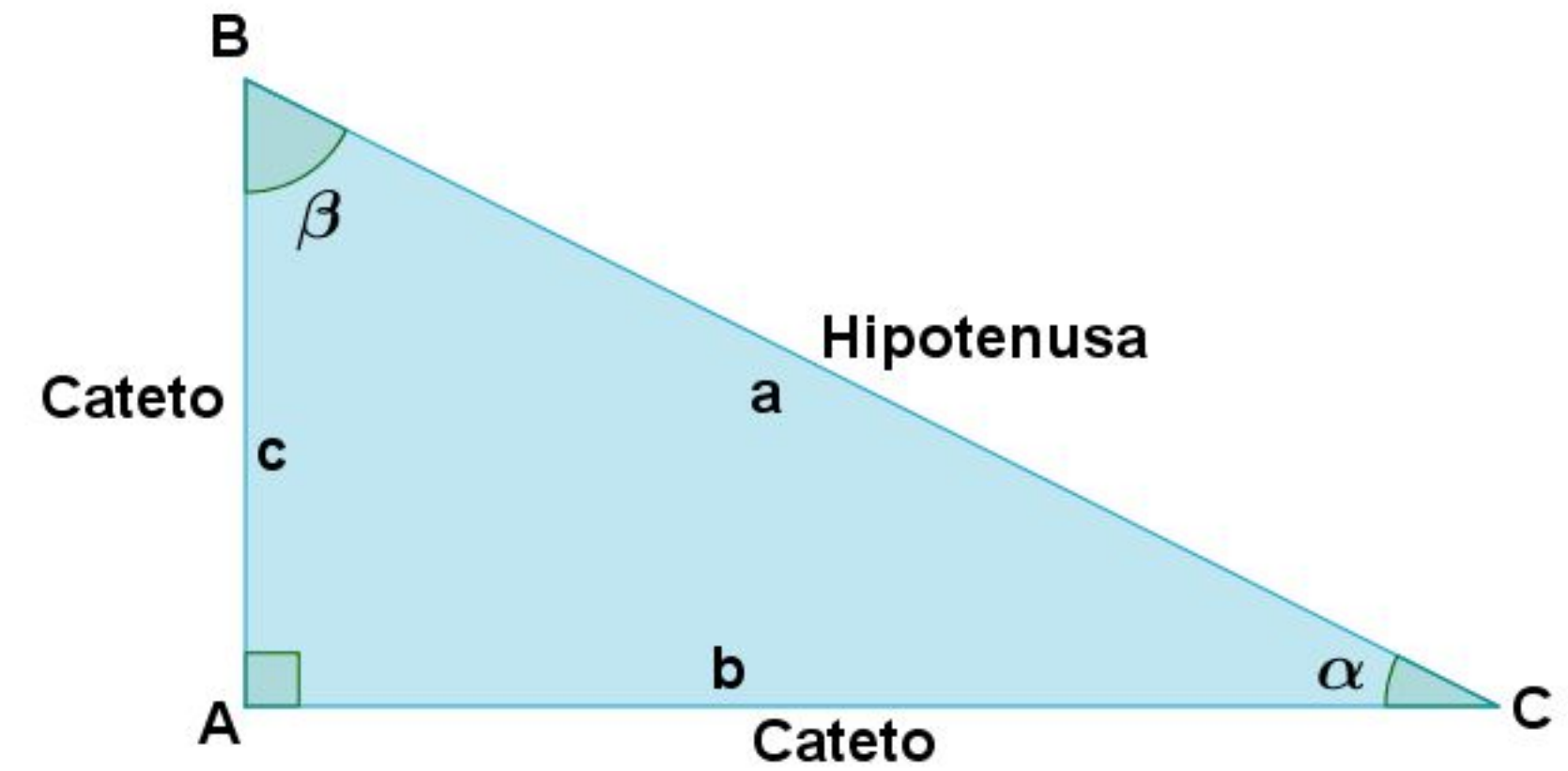


Mínimos locais

Tangente



$$\text{tg}(\hat{A}) = \frac{\text{cateto oposto}}{\text{cateto adjacente}}$$

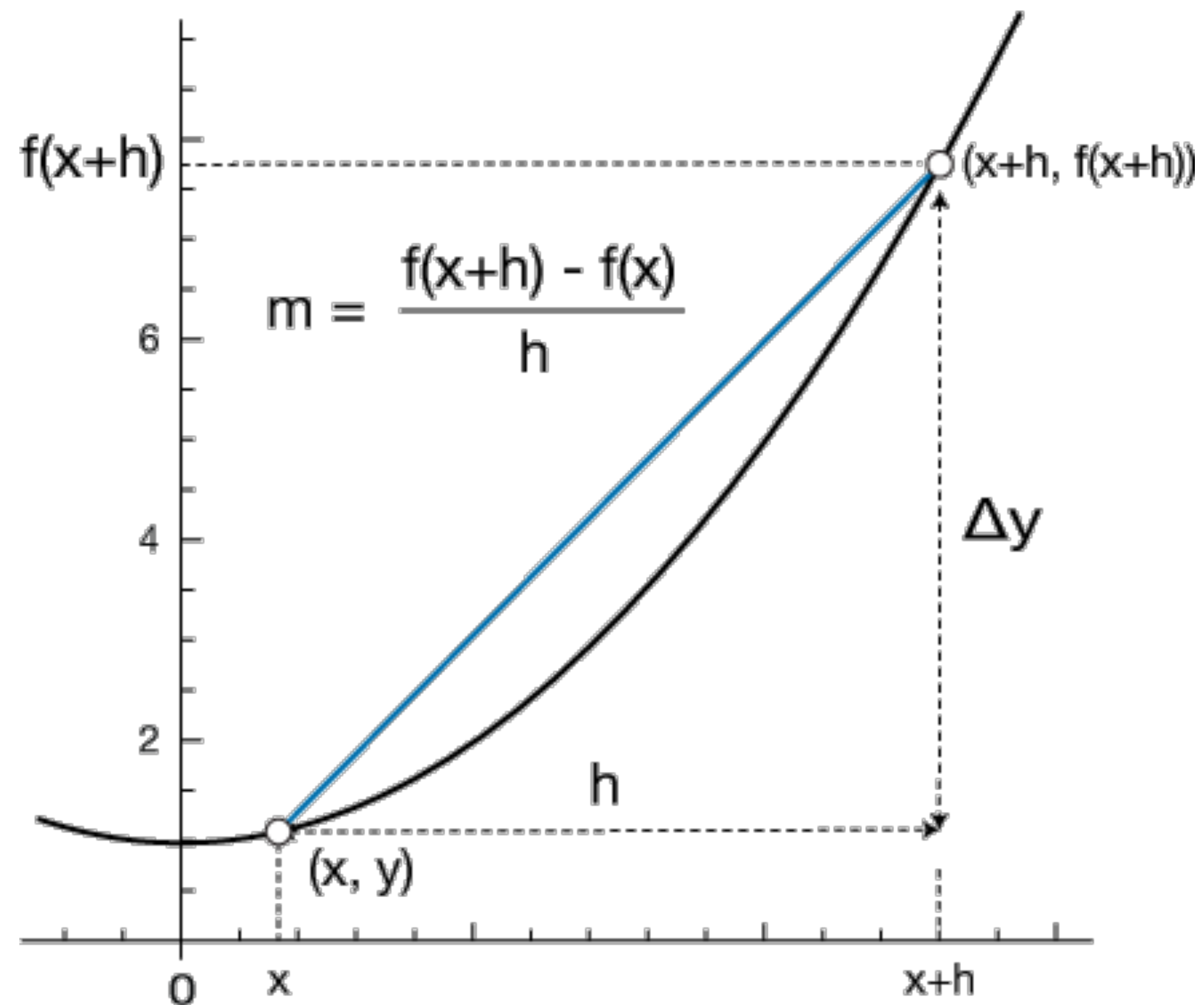


$$\text{tangente}(\alpha) = \frac{c}{b}$$

$$\text{tg}(\theta) = \frac{\text{sen}(\theta)}{\text{cos}(\theta)}$$

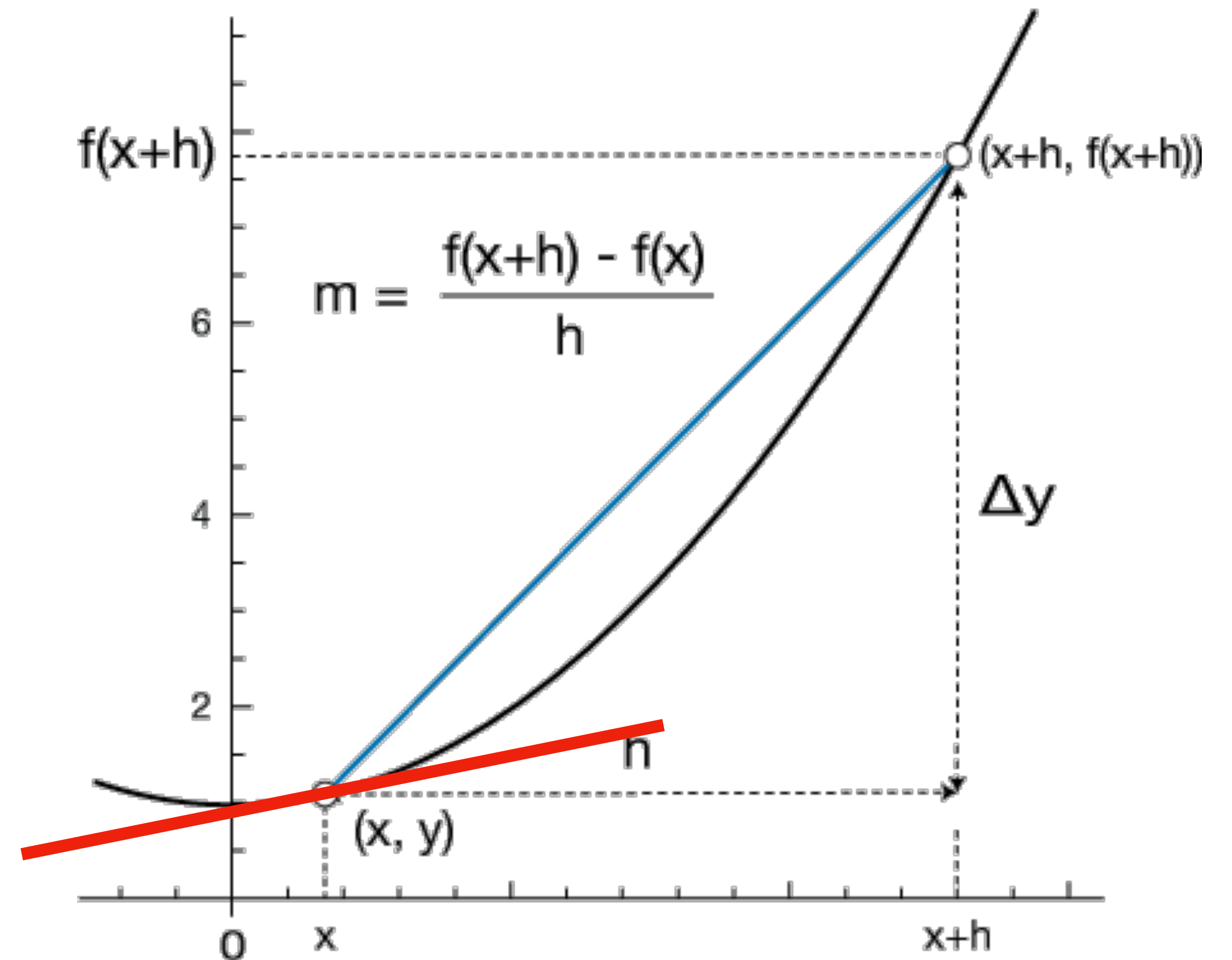
Máximo e mínimo de uma função

$$m = \frac{f(x+h) - f(x)}{h}$$



Máximo e mínimo de uma função

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

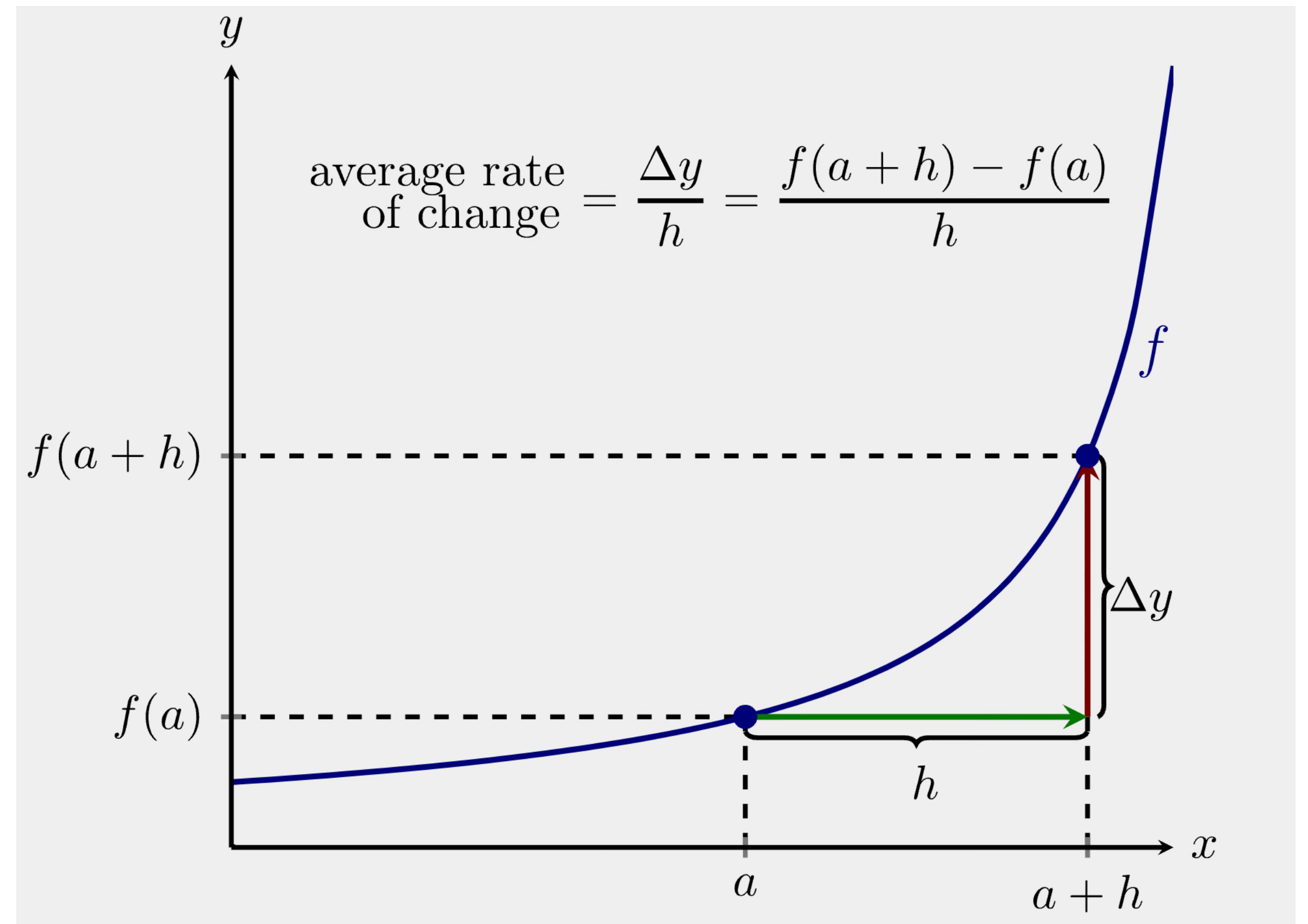


Máximo e mínimo de uma função

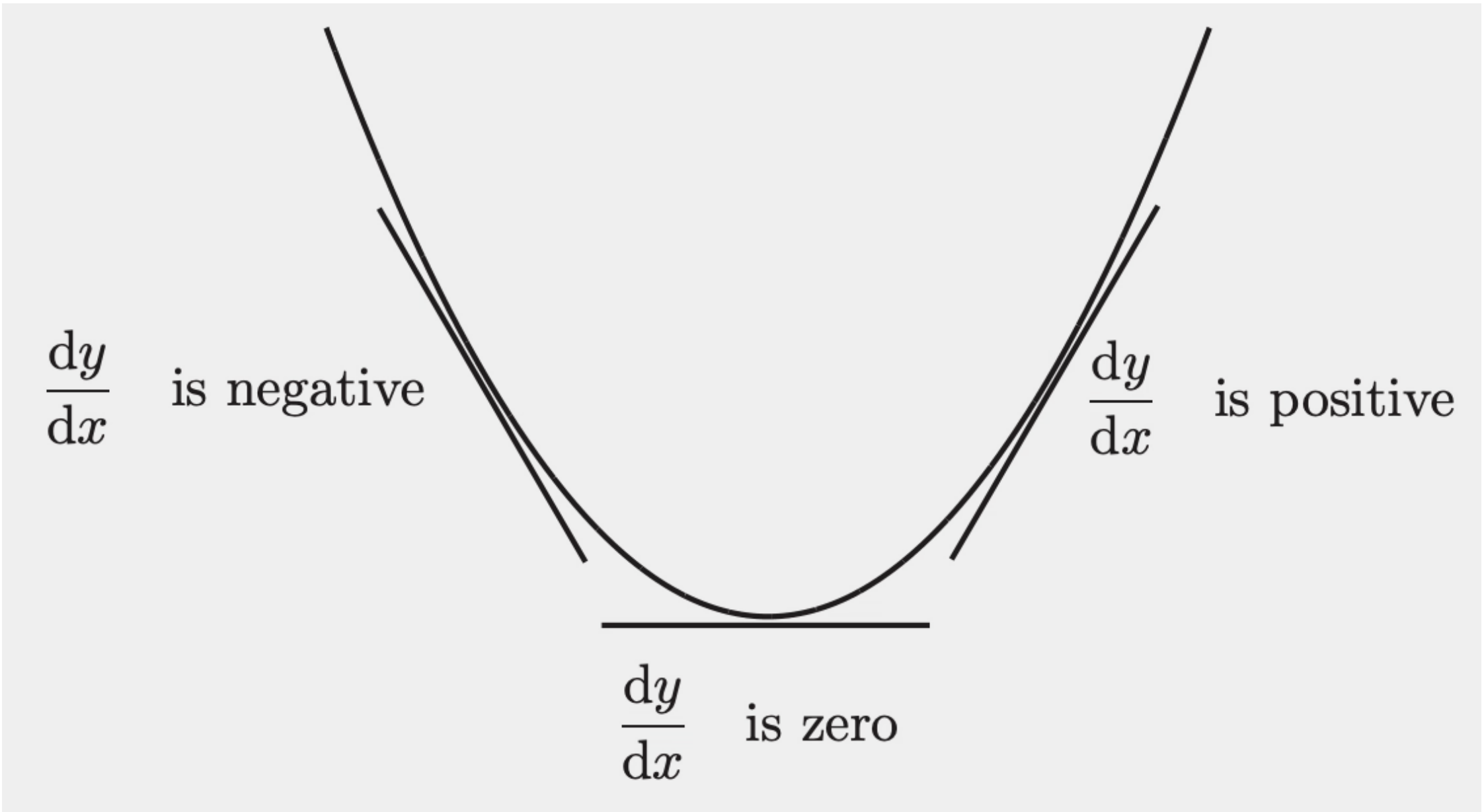
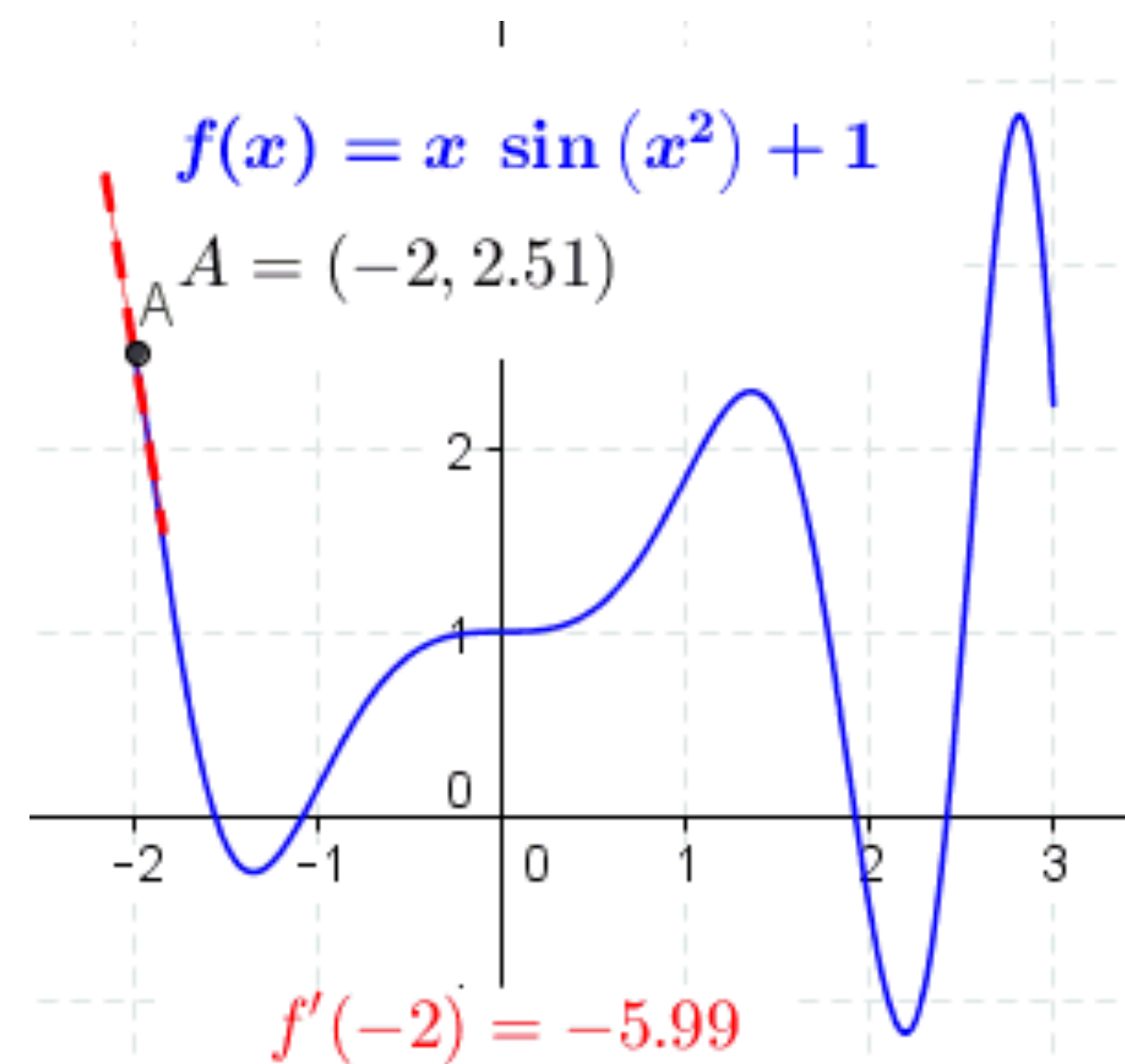
$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

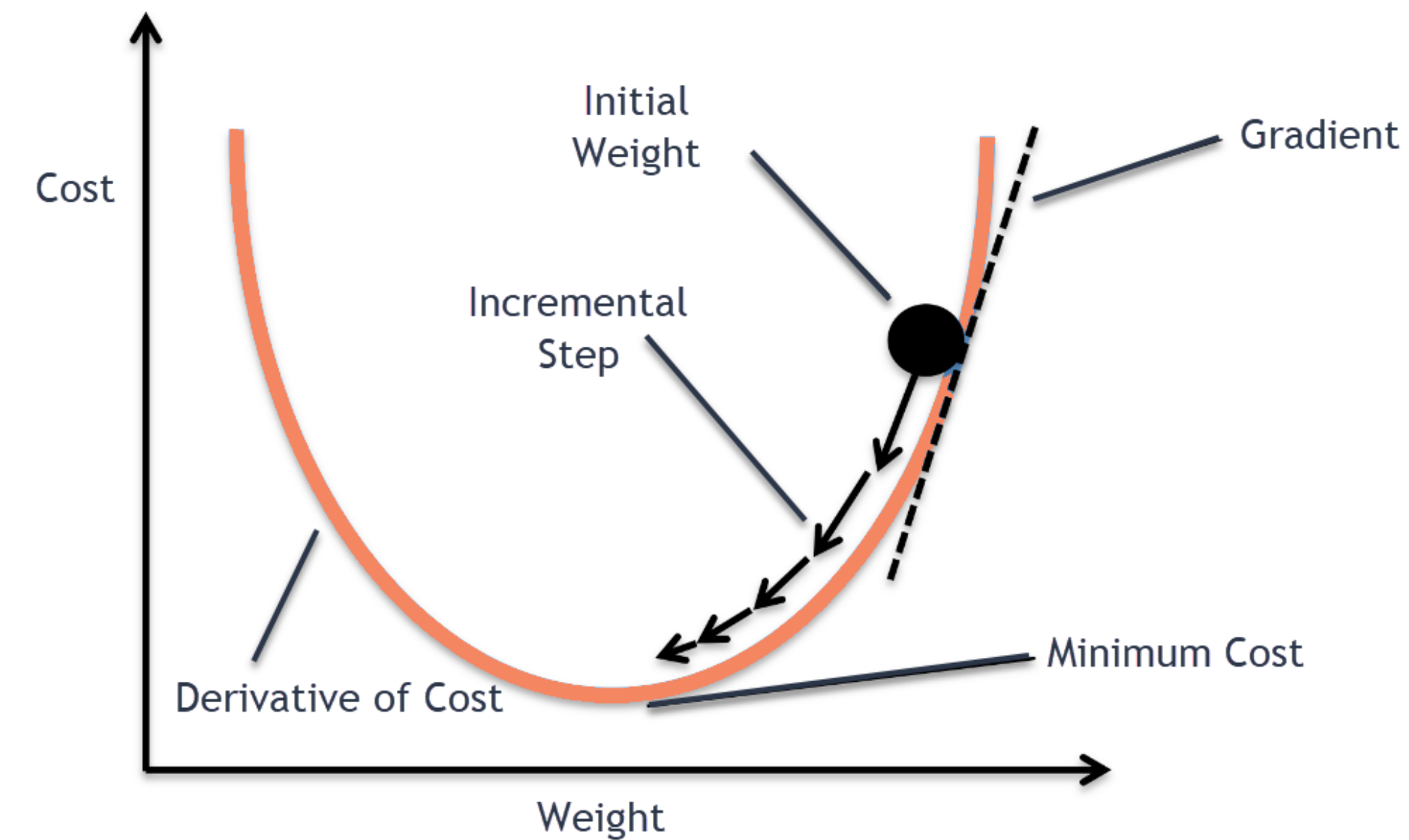
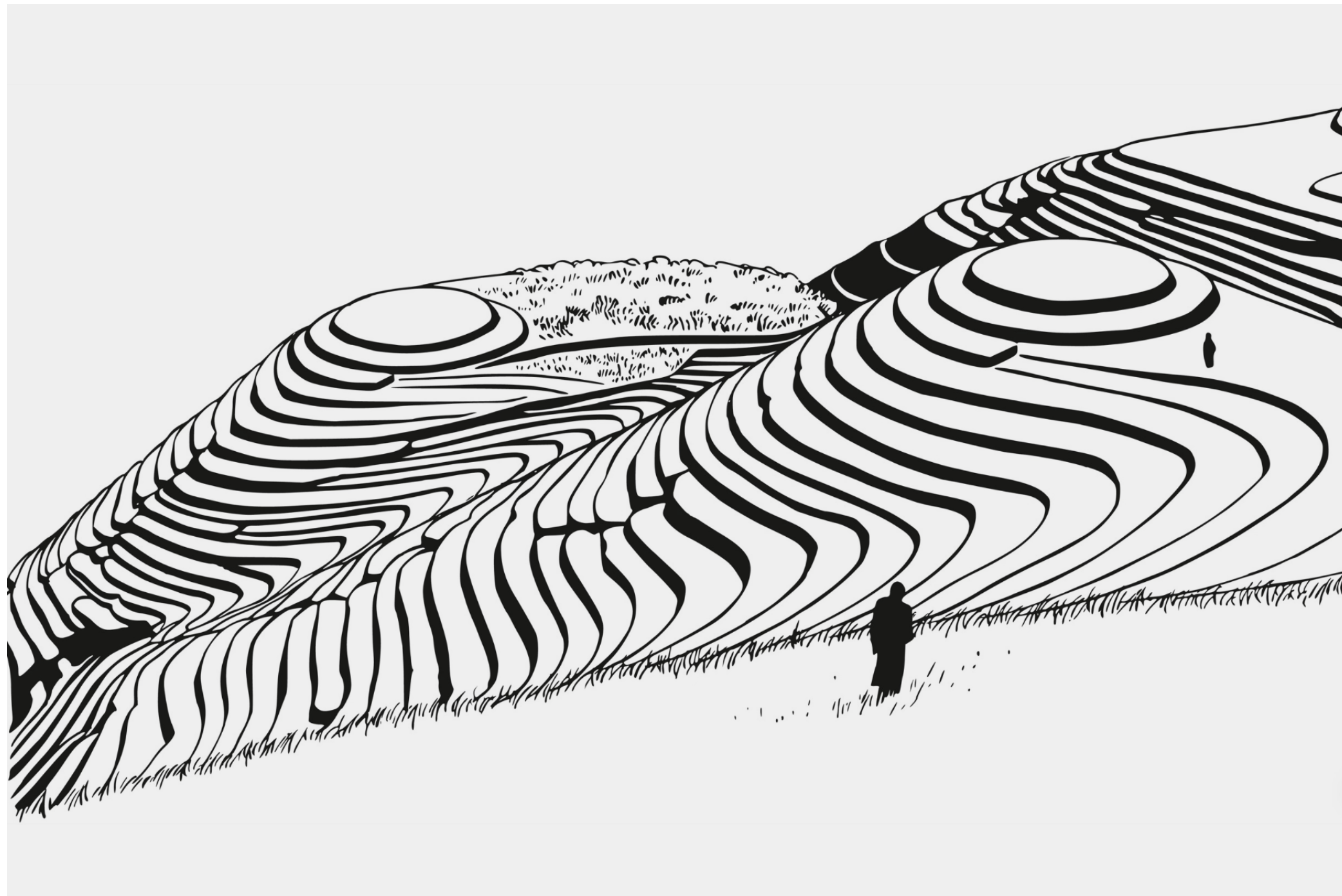
$$f'(a) = \frac{df}{dx}(a) = \left. \frac{df}{dx} \right|_{x=a}.$$



Máximo e mínimo de uma função

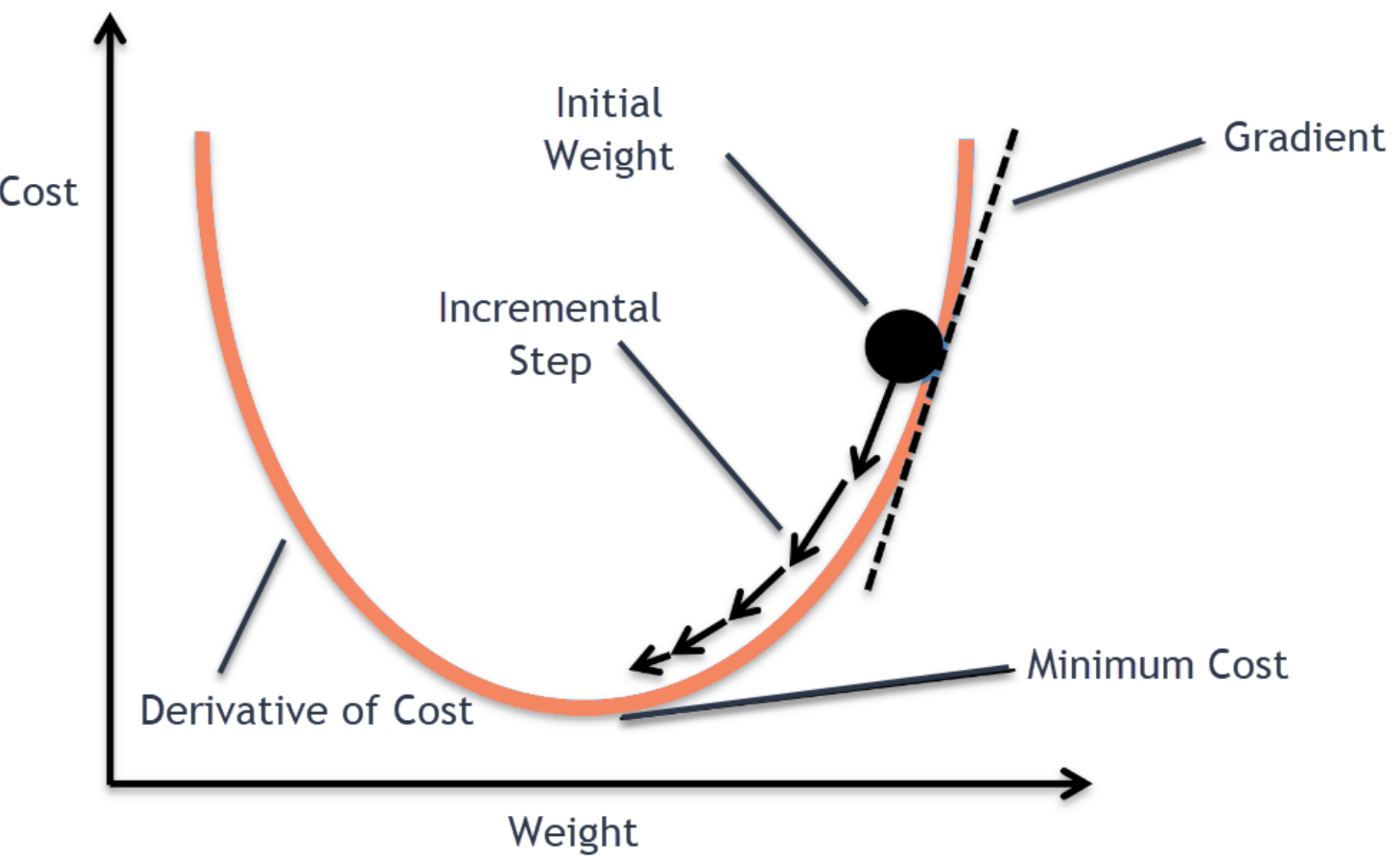
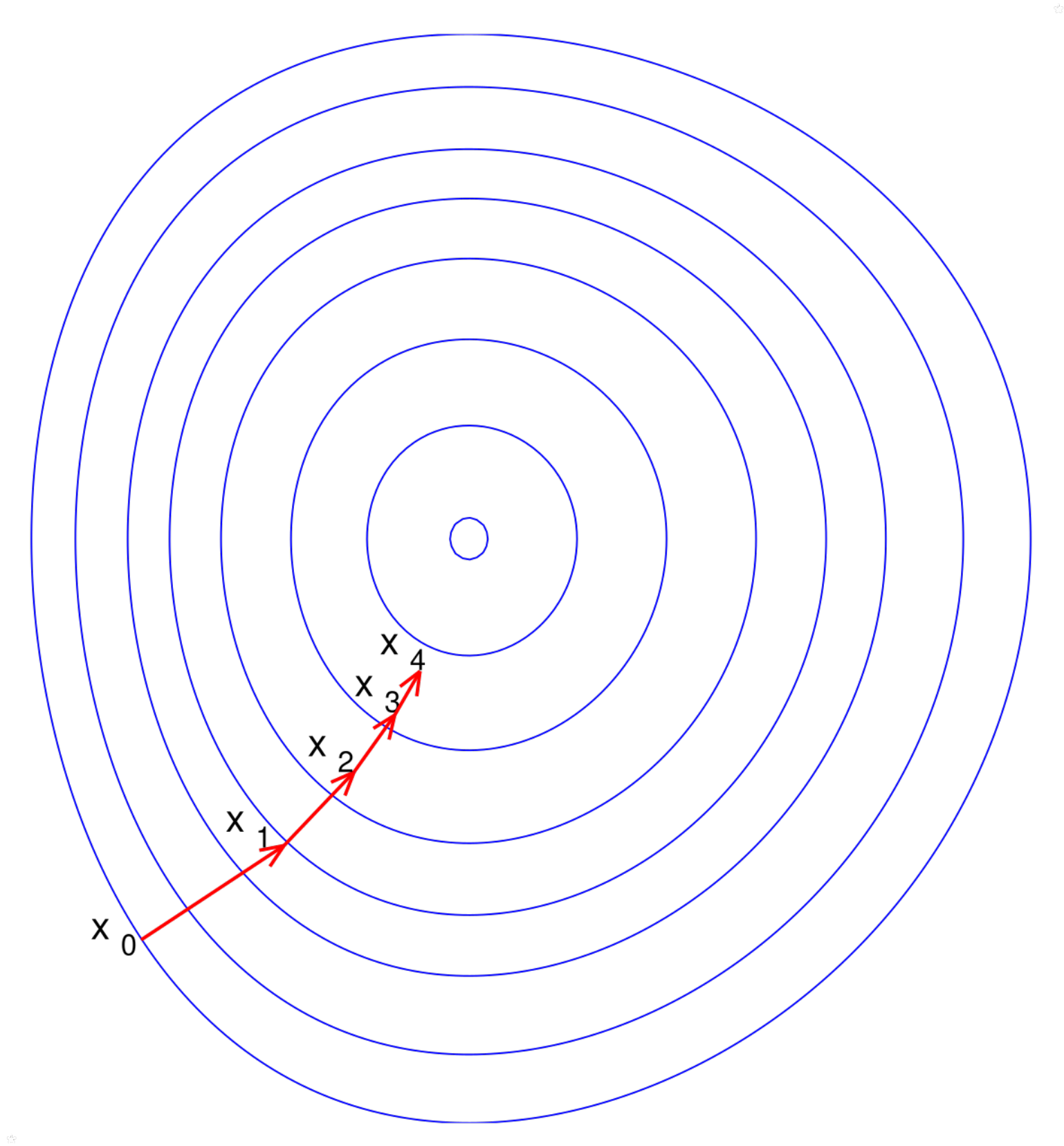


Descida do gradiente



$$\hat{\theta} = \arg \min_{\theta} \underbrace{\frac{1}{n} \sum_{i=1}^n \overbrace{L(\hat{y}(\mathbf{x}_i; \theta), y_i)}^{\text{loss function}}}_{\text{cost function } J(\theta)}.$$

Descida do gradiente

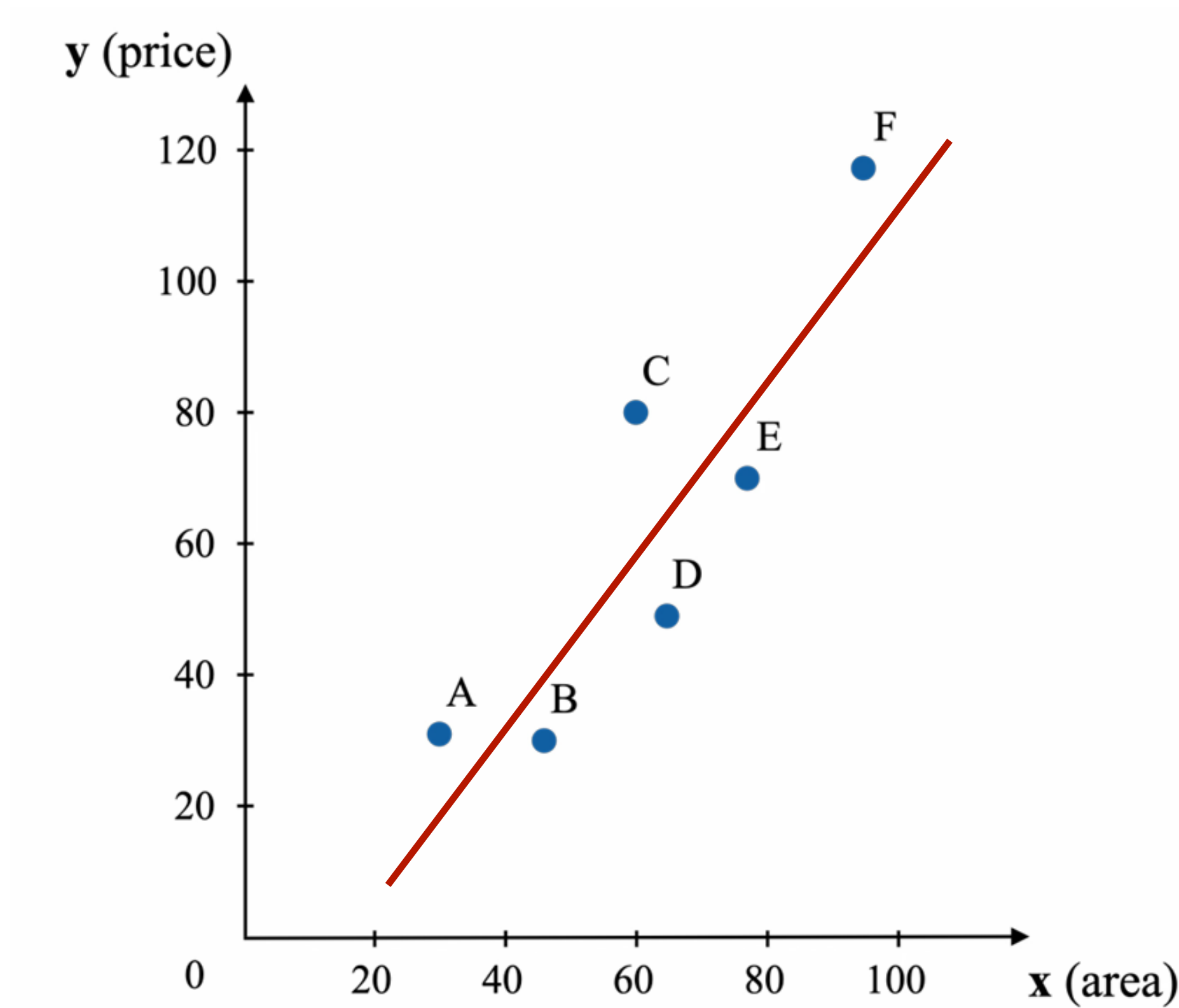


Descida do gradiente

Conjunto de dados:

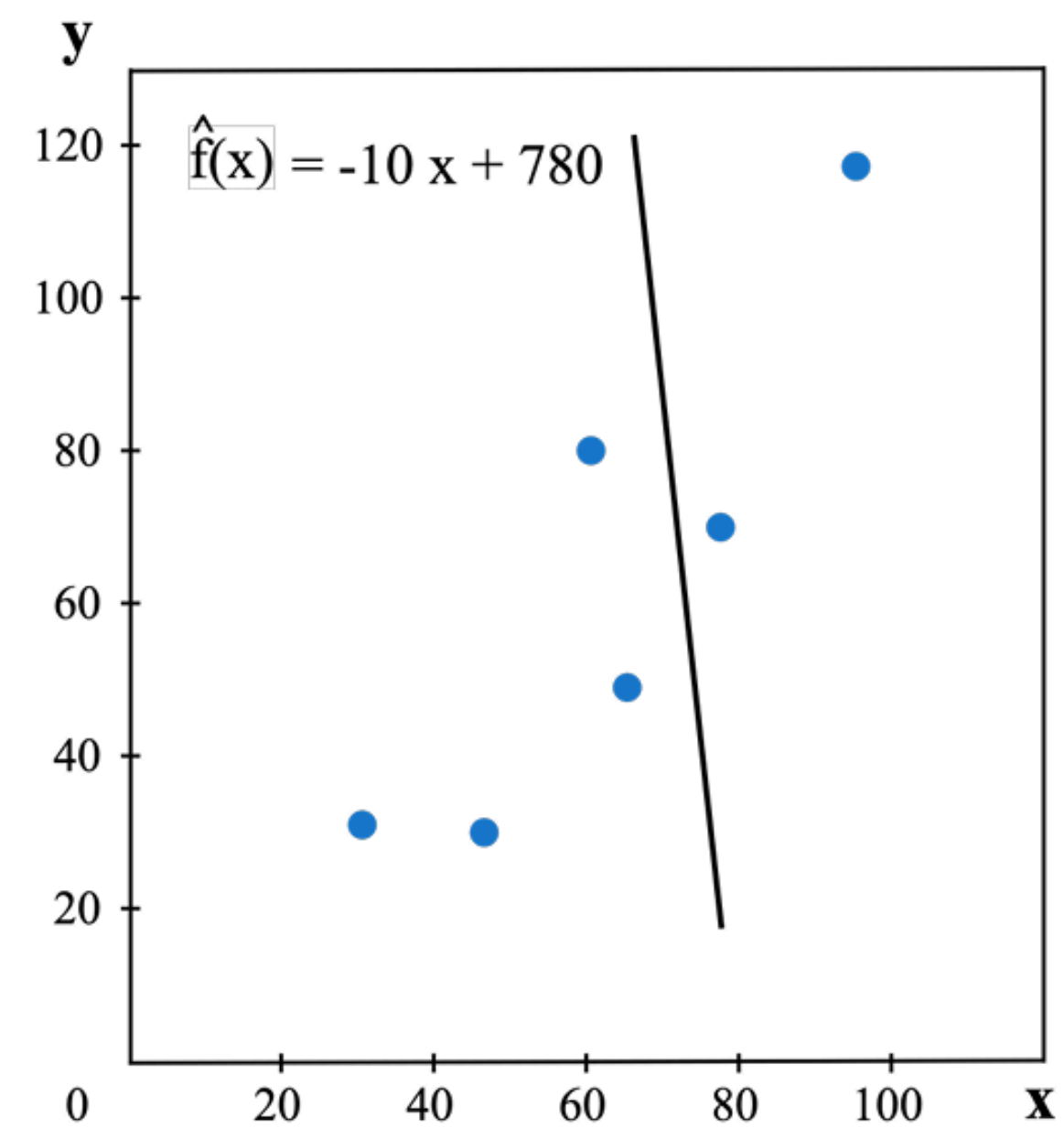
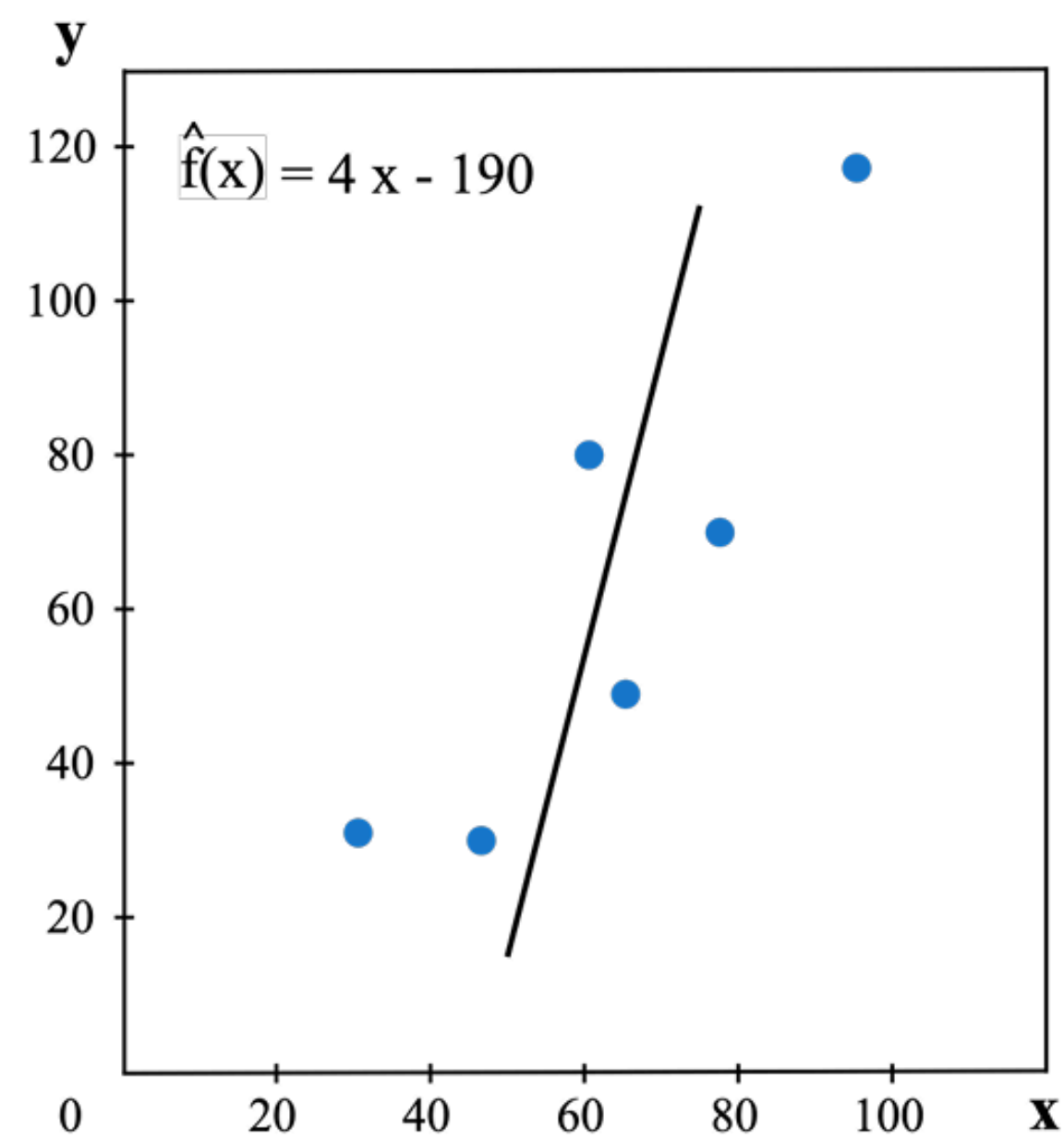
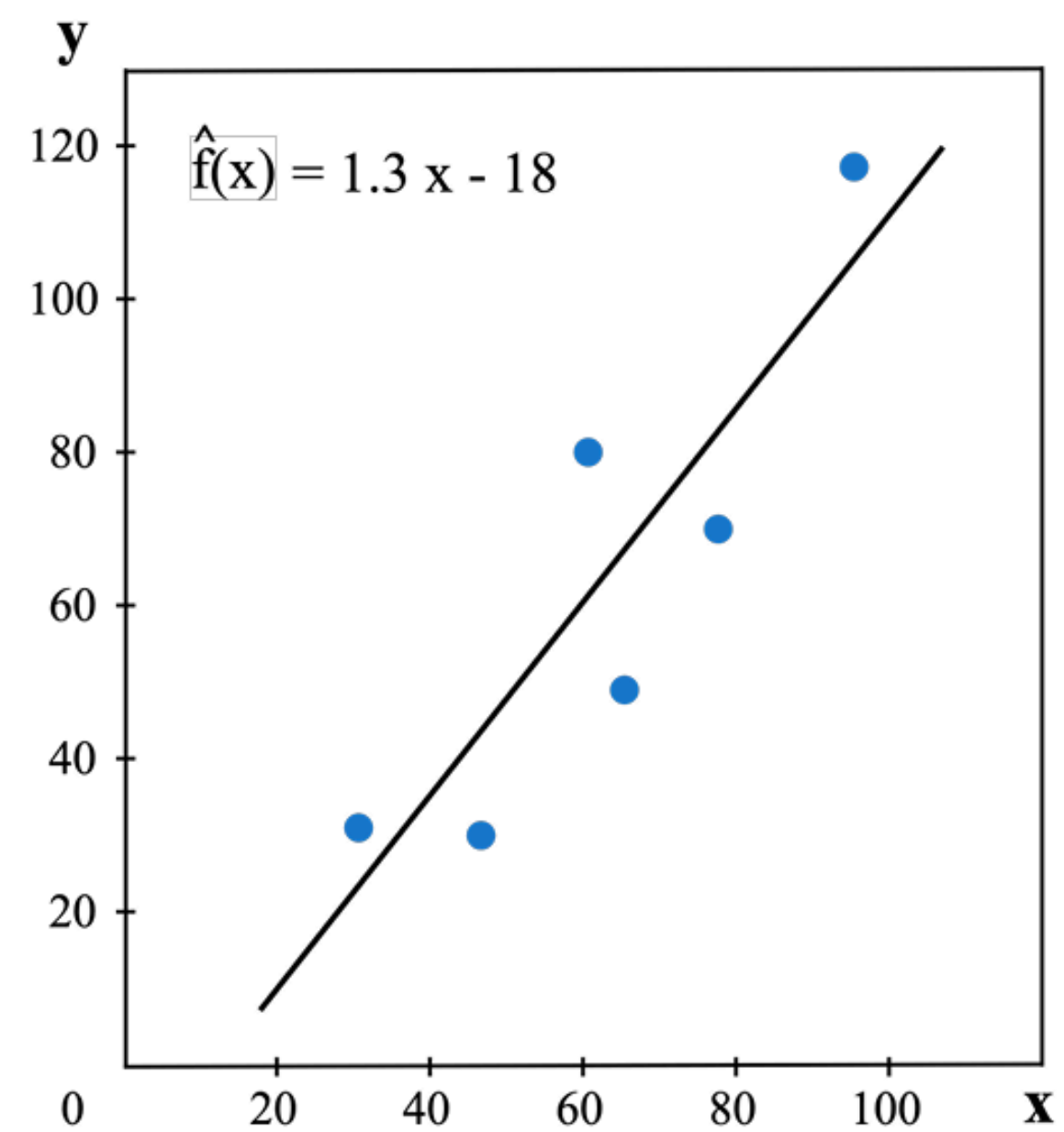
Amsterdam Apartments		
<i>n.</i>	Area (<i>m</i> ²)	Price (€10,000)
A	30	31
B	46	30
C	60	80
D	65	49
E	77	70
F	95	118

$$\hat{f}(x) = ax + b$$



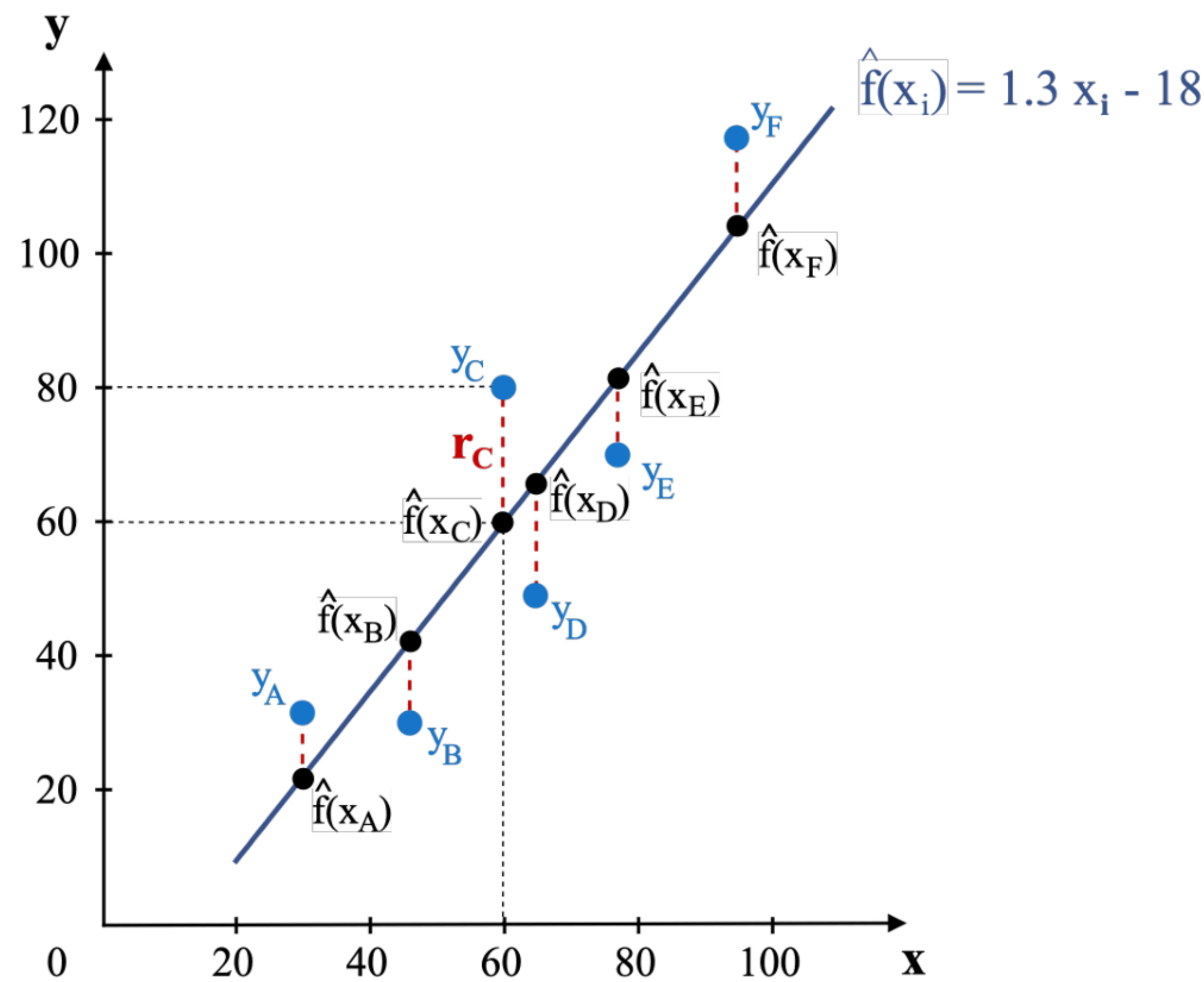
Descida do gradiente

$$\hat{f}(x) = ax + b$$

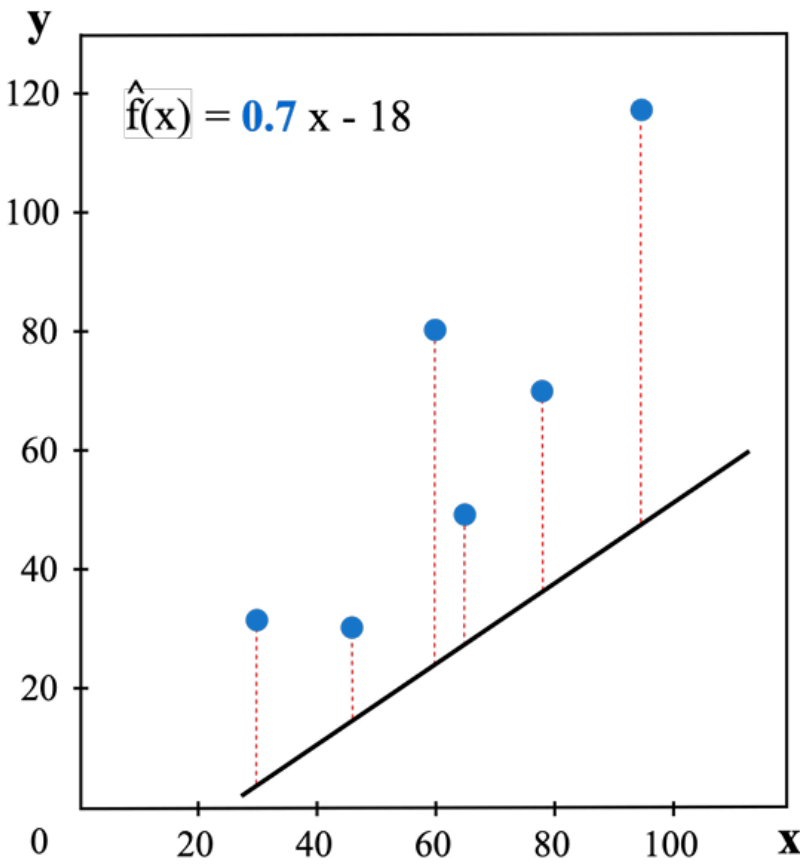


Descida do gradiente

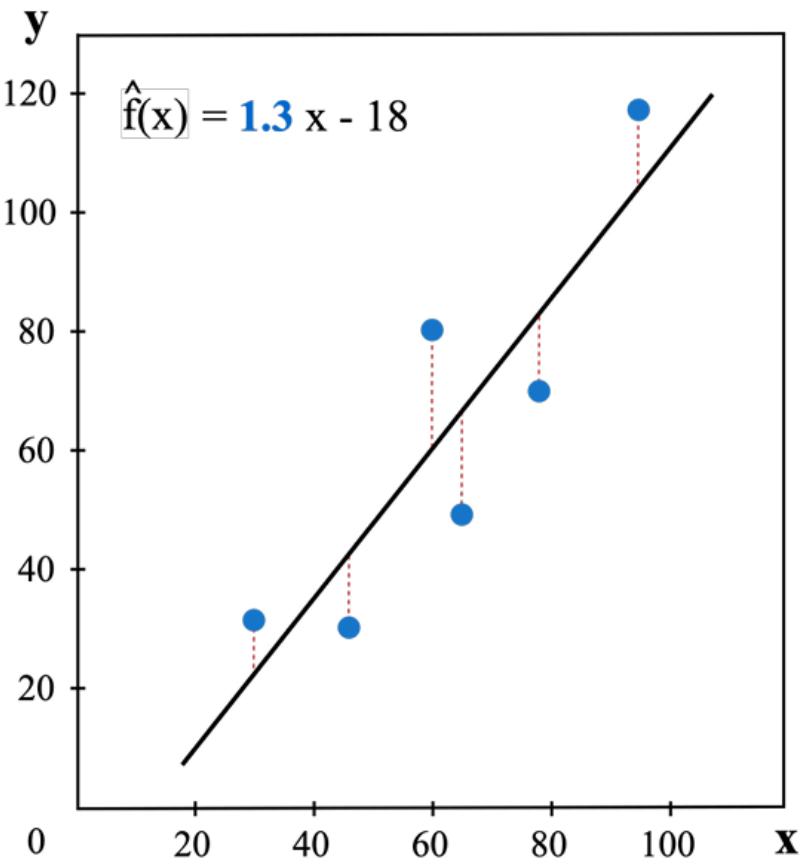
$$\hat{f}(x) = ax + b$$



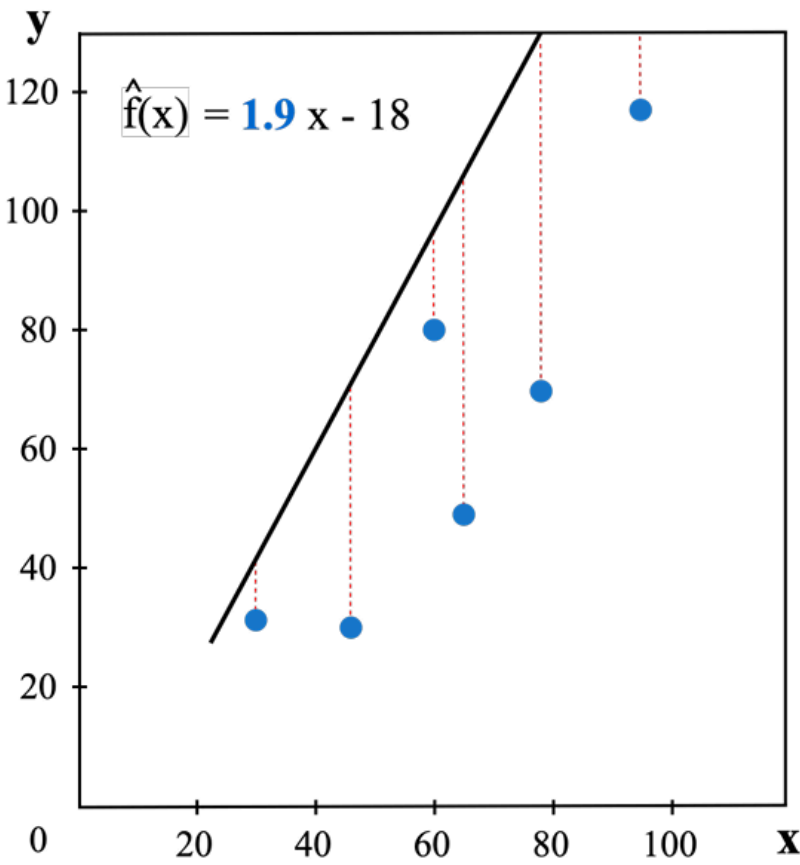
Descida do gradiente



(a) $a = 0.7$; $SSR = 10,624.95$.

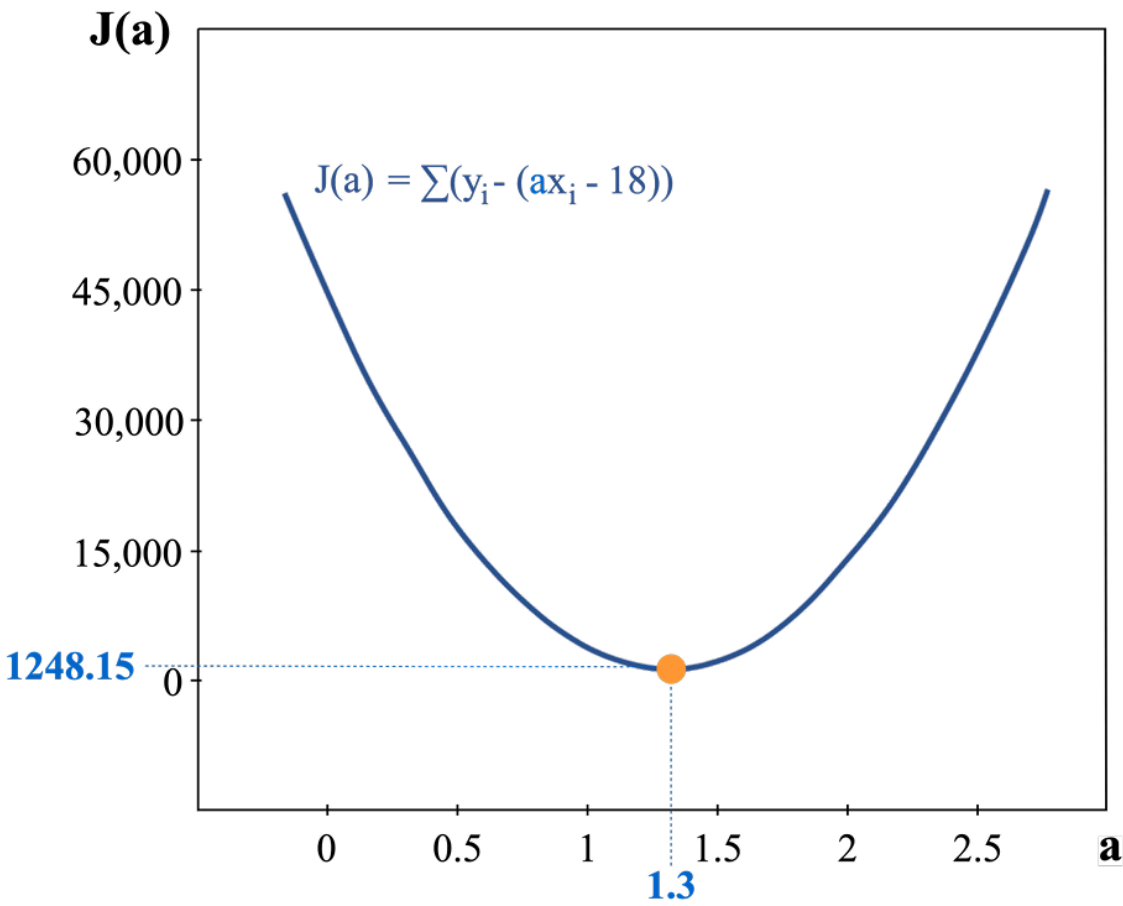
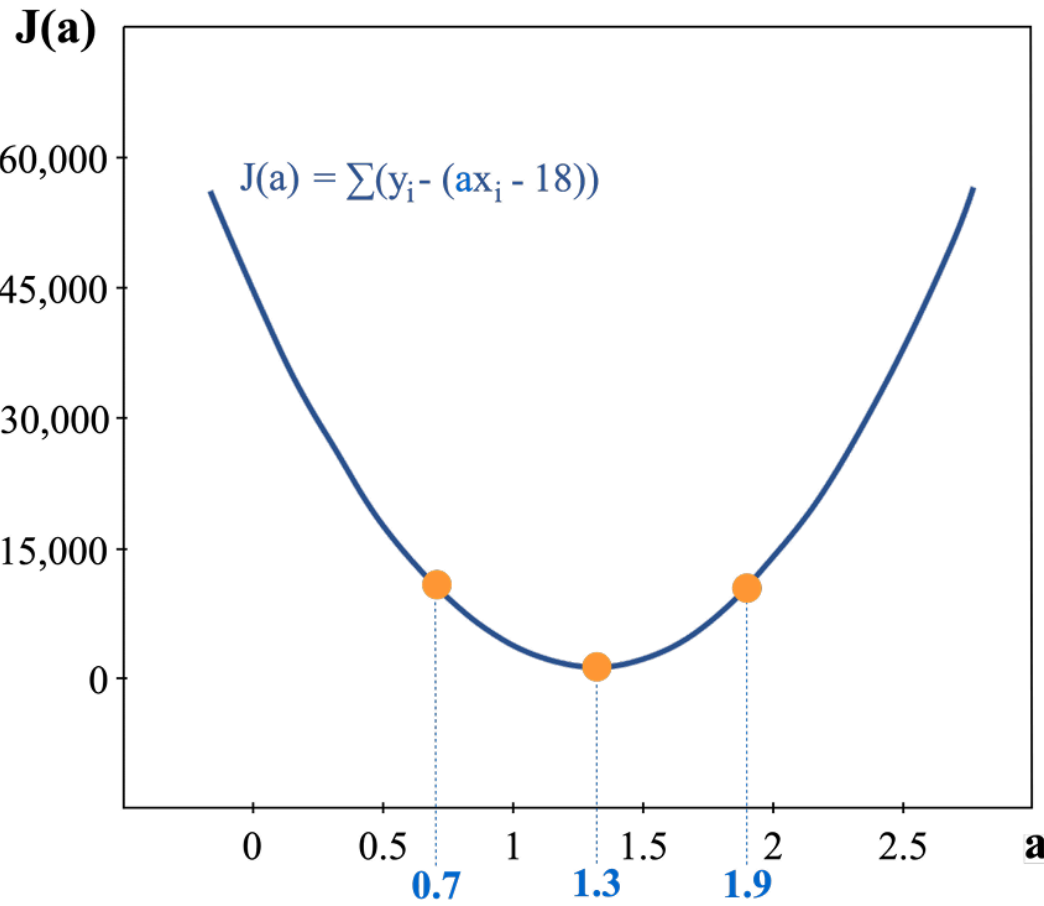


(b) $a = 1.3$; $SSR = 1,248.15$.



(c) $a = 1.9$; $SSR = 10,443.75$.

$$J(a) = \sum_{i=1}^n (y_i - (ax_i - 18))^2$$



Descida do gradiente

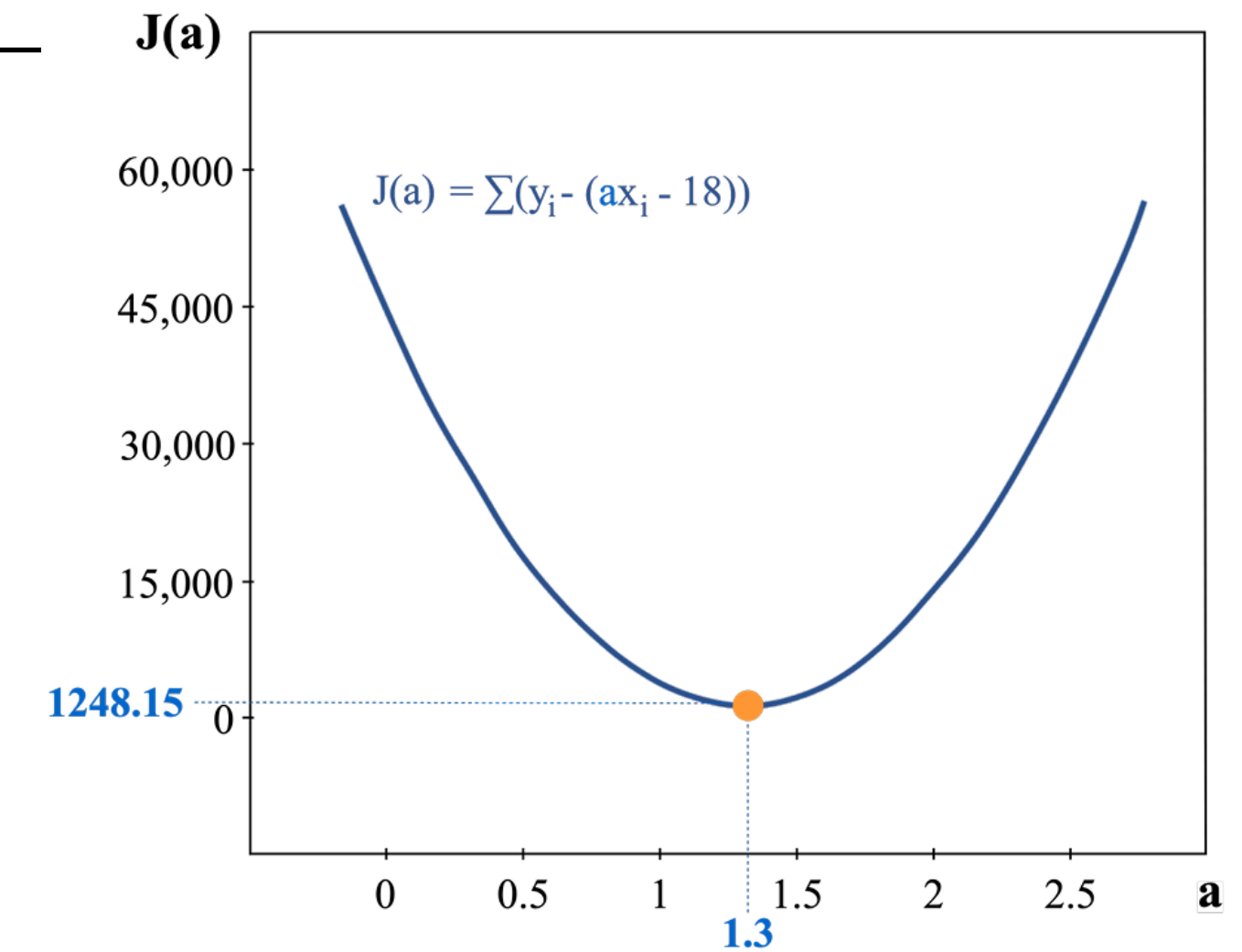
Primeira iteração: $a_0 = 0$

$$a^* = \arg \min_a J(a)$$

$$= \arg \min_a \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

$$J(a) = \sum_{i=1}^n (y_i - (ax_i - 18))^2$$

$$\begin{aligned} \frac{\partial J}{\partial a} &= \frac{\partial}{\partial a} \sum_{i=1}^n \left(y_i - (ax_i - 18) \right)^2 \\ &= \sum_{i=1}^n \frac{\partial}{\partial a} \left(y_i - (ax_i - 18) \right)^2 \\ &= \sum_{i=1}^n -2 \cdot x_i \cdot \left(y_i - (ax_i - 18) \right) \end{aligned}$$



Descida do gradiente

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F	95	118

$$J(a) = \sum_{i=1}^n \left(y_i - (ax_i - 18) \right)^2$$

$$\begin{aligned} J(a) = & \left(31 - (a \cdot 30 - 18) \right)^2 + \left(30 - (a \cdot 46 - 18) \right)^2 \\ & + \left(80 - (a \cdot 60 - 18) \right)^2 + \left(49 - (a \cdot 65 - 18) \right)^2 \\ & + \left(70 - (a \cdot 77 - 18) \right)^2 + \left(118 - (a \cdot 95 - 18) \right)^2 \end{aligned}$$

$$\begin{aligned} \frac{\partial J}{\partial a} = & \frac{\partial}{\partial a} \left(31 - (a \cdot 30 - 18) \right)^2 + \frac{\partial}{\partial a} \left(30 - (a \cdot 46 - 18) \right)^2 \\ & + \frac{\partial}{\partial a} \left(80 - (a \cdot 60 - 18) \right)^2 + \frac{\partial}{\partial a} \left(49 - (a \cdot 65 - 18) \right)^2 \\ & + \frac{\partial}{\partial a} \left(70 - (a \cdot 77 - 18) \right)^2 + \frac{\partial}{\partial a} \left(118 - (a \cdot 95 - 18) \right)^2 \end{aligned}$$

Descida do gradiente

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$$\begin{aligned} \frac{\partial J}{\partial a} = & (-2 \cdot 30) \cdot \left(31 - (a \cdot 30 - 18) \right) + (-2 \cdot 46) \cdot \left(30 - (a \cdot 46 - 18) \right) \\ & + (-2 \cdot 60) \cdot \left(80 - (a \cdot 60 - 18) \right) + (-2 \cdot 65) \cdot \left(49 - (a \cdot 65 - 18) \right) \\ & + (-2 \cdot 77) \cdot \left(70 - (a \cdot 77 - 18) \right) + (-2 \cdot 95) \cdot \left(118 - (a \cdot 95 - 18) \right) \end{aligned}$$

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$$a_0 = 0$$

$$a_{new} = a - \underbrace{\frac{\partial J}{\partial a} \cdot l}_{\text{Step Size}}$$

$$\begin{aligned} \frac{\partial J}{\partial a} = & (-2 \cdot 30) \cdot (31 - (\mathbf{0} \cdot 30 - 18)) + (-2 \cdot 46) \cdot (30 - (\mathbf{0} \cdot 46 - 18)) \\ & + (-2 \cdot 60) \cdot (80 - (\mathbf{0} \cdot 60 - 18)) + (-2 \cdot 65) \cdot (49 - (\mathbf{0} \cdot 65 - 18)) \\ & + (-2 \cdot 77) \cdot (70 - (\mathbf{0} \cdot 77 - 18)) + (-2 \cdot 95) \cdot (118 - (\mathbf{0} \cdot 95 - 18)) \\ = & -67218 \end{aligned}$$

Descida do gradiente

Amsterdam Apartments		
<i>n.</i>	Area (<i>m</i> ²)	Price (€10,000)
A	30	31
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D	65	49
E	77	70
F	95	118

$$J(a) = \sum_{i=1}^n \left(y_i - (ax_i - 18) \right)^2$$

$$a_0 = 0$$

↓

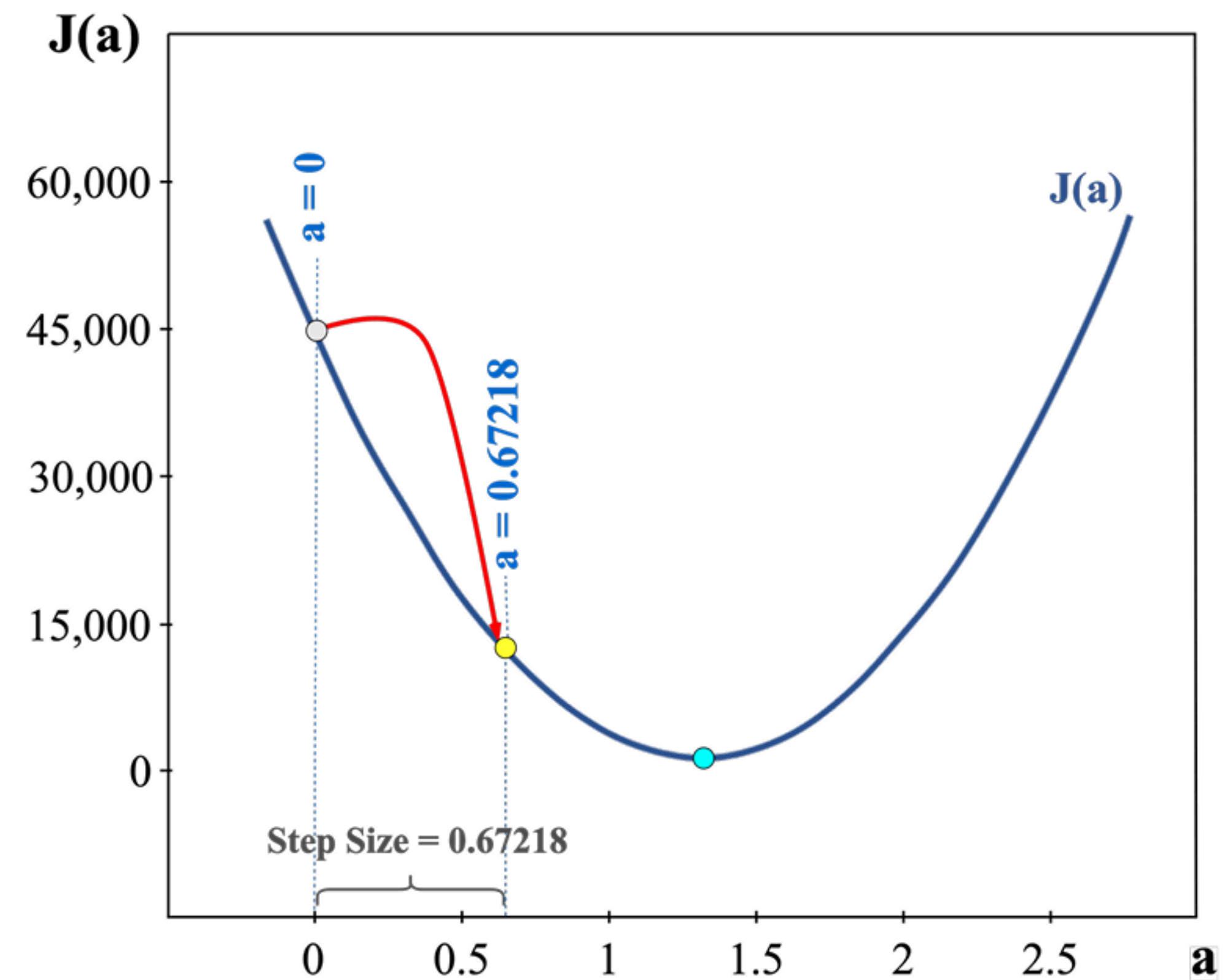
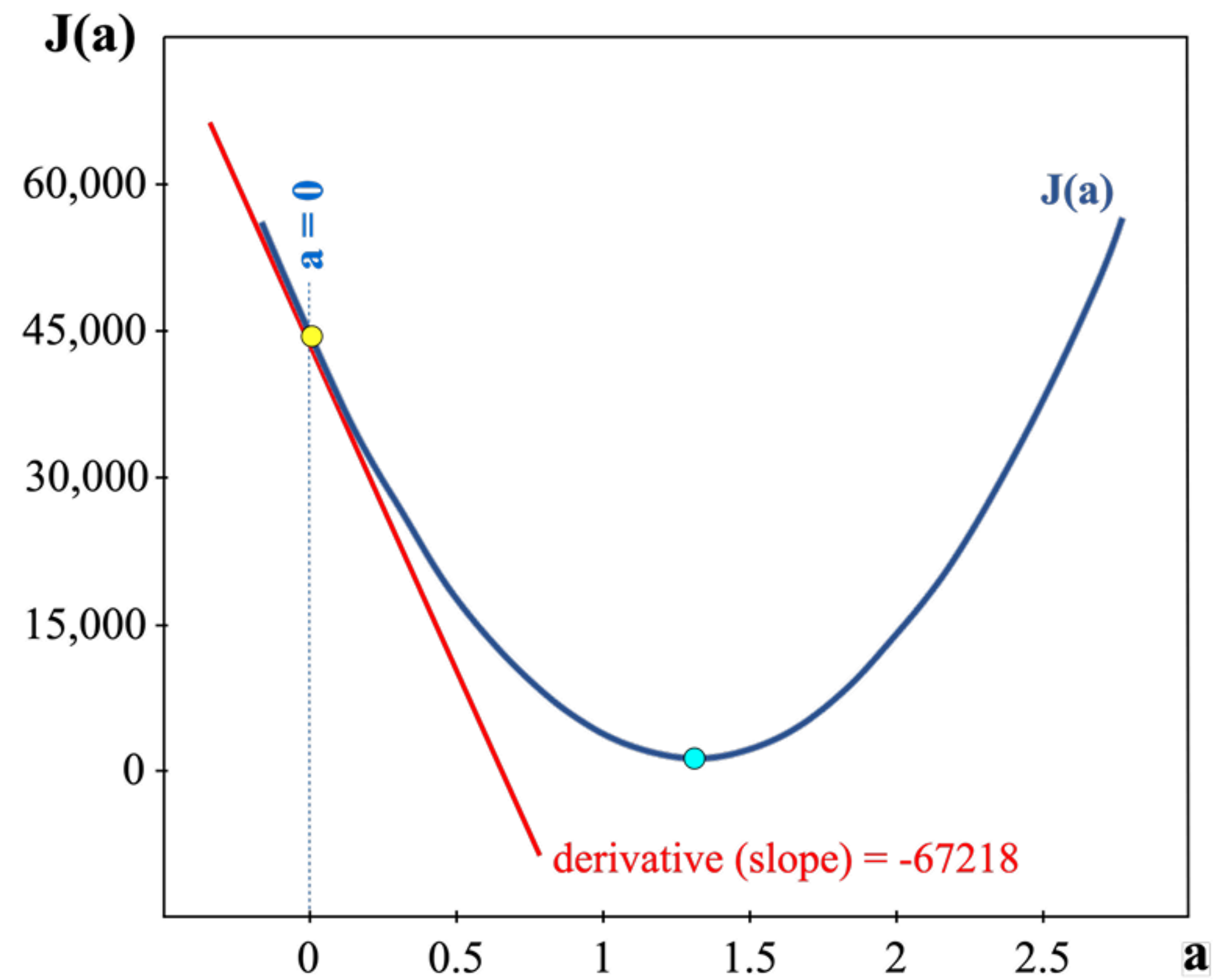
$$a_{new} = a - \underbrace{\frac{\partial J}{\partial a} \cdot l}_{\text{Step Size}}$$

$$\begin{aligned} a_{new} &= 0 - \underbrace{(-67218) \cdot 0.00001}_{\text{Step Size}} \\ &= 0 - \underbrace{(-0.67218)}_{\text{Step Size}} \\ &= 0.67218 \end{aligned}$$

Descida do gradiente

Primeira iteração: $a_0 = 0$

$$a_0 = 0 \longrightarrow a_1 = 0.67218$$



Descida do gradiente

Segunda iteração: $a_1 = 0.67218$

$$J(a) = \sum_{i=1}^n (y_i - (ax_i - 18))^2$$

$$\begin{aligned} \frac{\partial J}{\partial a} = & (-2 \cdot 30) \cdot (31 - (a \cdot 30 - 18)) + (-2 \cdot 46) \cdot (30 - (a \cdot 46 - 18)) \\ & + (-2 \cdot 60) \cdot (80 - (a \cdot 60 - 18)) + (-2 \cdot 65) \cdot (49 - (a \cdot 65 - 18)) \\ & + (-2 \cdot 77) \cdot (70 - (a \cdot 77 - 18)) + (-2 \cdot 95) \cdot (118 - (a \cdot 95 - 18)) \end{aligned}$$

$$a_{new} = a - \underbrace{\frac{\partial J}{\partial a} \cdot l}_{\text{Step Size}}$$

Descida do gradiente

Segunda iteração: $a_1 = 0.67218$

$$J(a) = \sum_{i=1}^n (y_i - (ax_i - 18))^2$$

$$\begin{aligned} \frac{\partial J}{\partial a} &= (-2 \cdot 30) \cdot (31 - (\mathbf{0.67218} \cdot 30 - 18)) + (-2 \cdot 46) \cdot (30 - (\mathbf{0.67218} \cdot 46 - 18)) \\ &\quad + (-2 \cdot 60) \cdot (80 - (\mathbf{0.67218} \cdot 60 - 18)) + (-2 \cdot 65) \cdot (49 - (\mathbf{0.67218} \cdot 65 - 18)) \\ &\quad + (-2 \cdot 77) \cdot (70 - (\mathbf{0.67218} \cdot 77 - 18)) + (-2 \cdot 95) \cdot (118 - (\mathbf{0.67218} \cdot 95 - 18)) \\ &= -32540.2338 \end{aligned}$$

Descida do gradiente

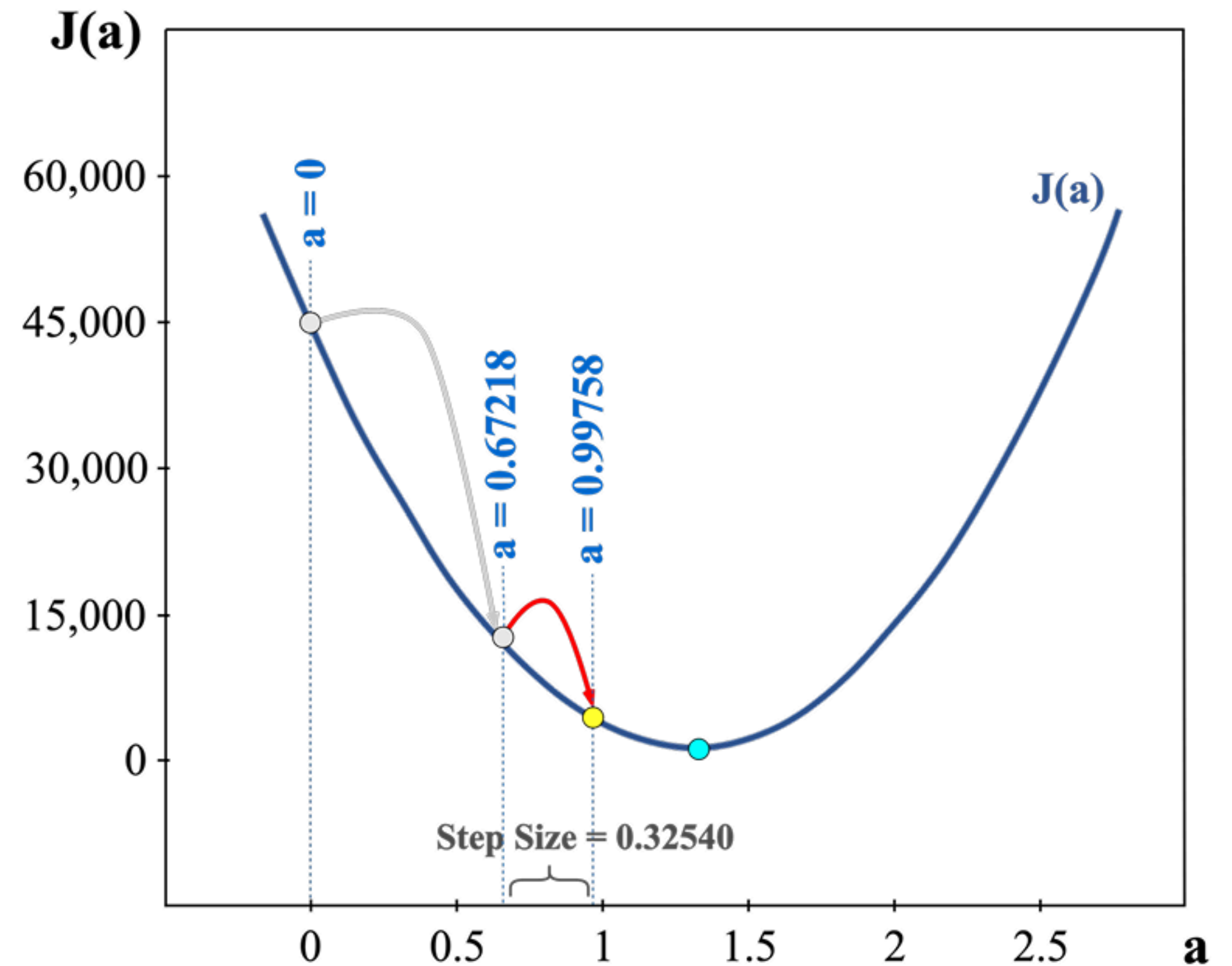
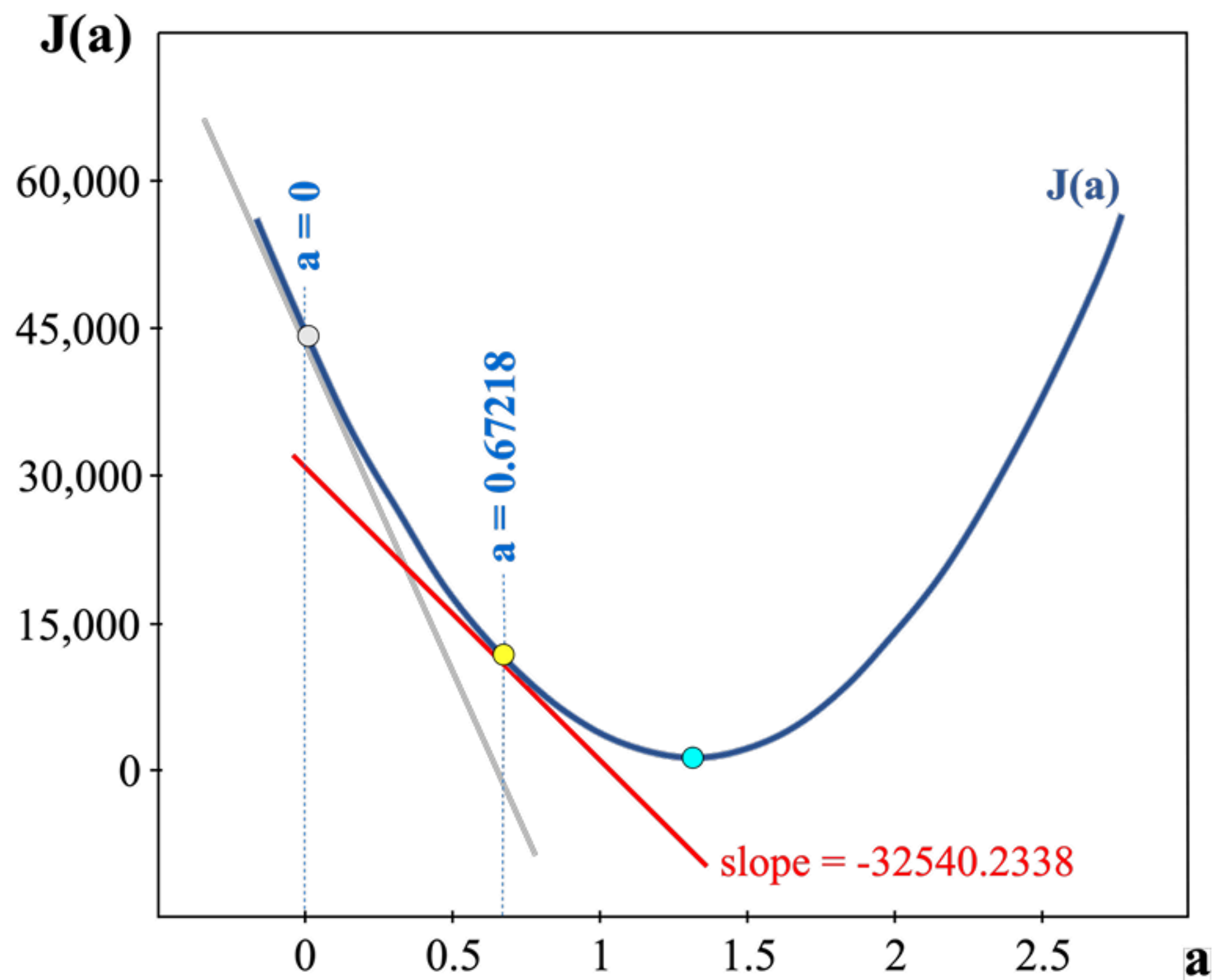
Segunda iteração: $a_1 = 0.67218$

$$\begin{aligned} a_{new} &= 0.67218 - \underbrace{(-32540.2338) \cdot 0.00001}_{\text{Step Size}} \\ &= 0.67218 - \underbrace{(-0.32540)}_{\text{Step Size}} \\ &= 0.99758 \end{aligned}$$

Descida do gradiente

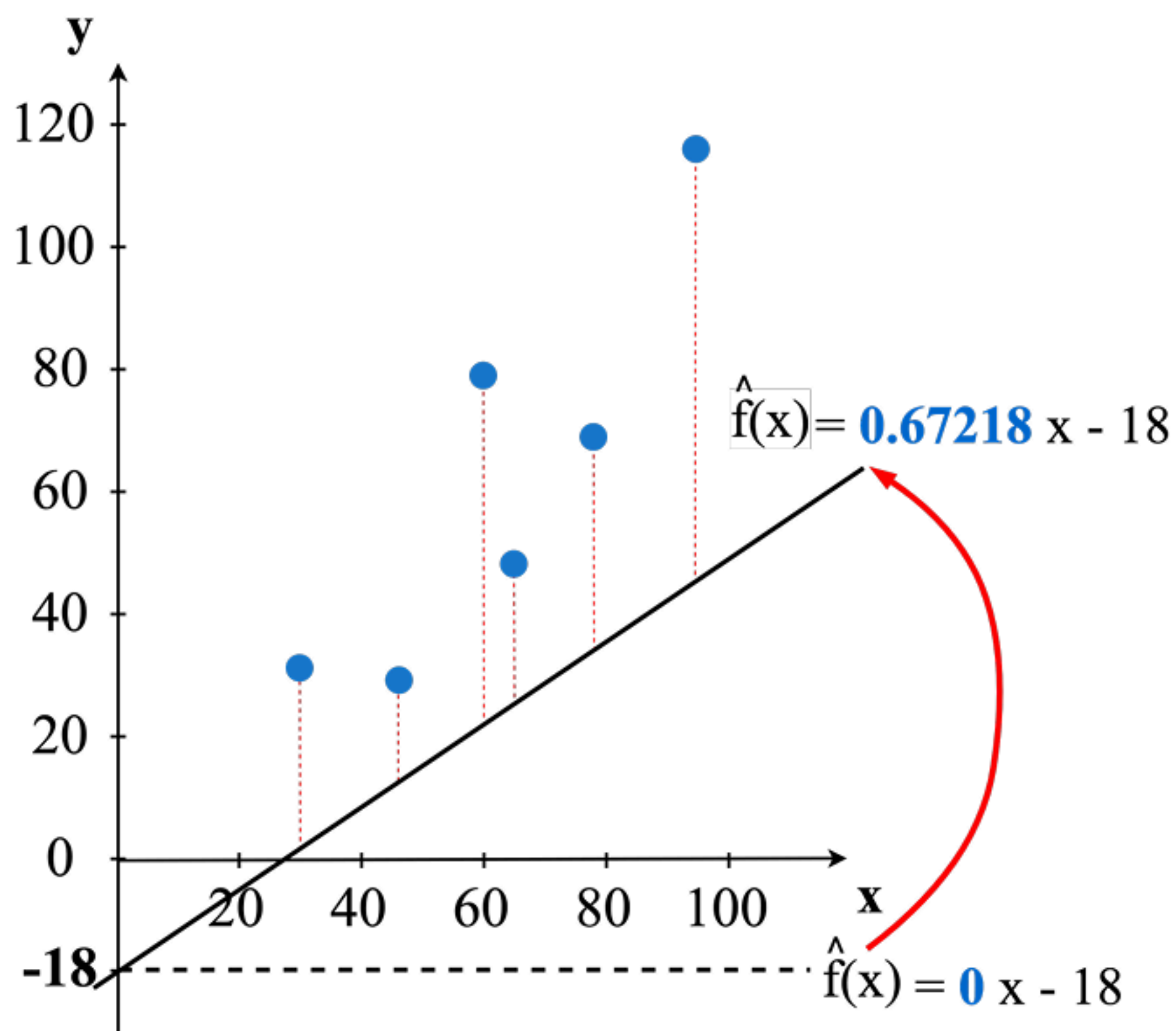
Segunda iteração: $a_1 = 0.67218$

$a_1 = 0.67218 \longrightarrow a_2 = 0.99758$

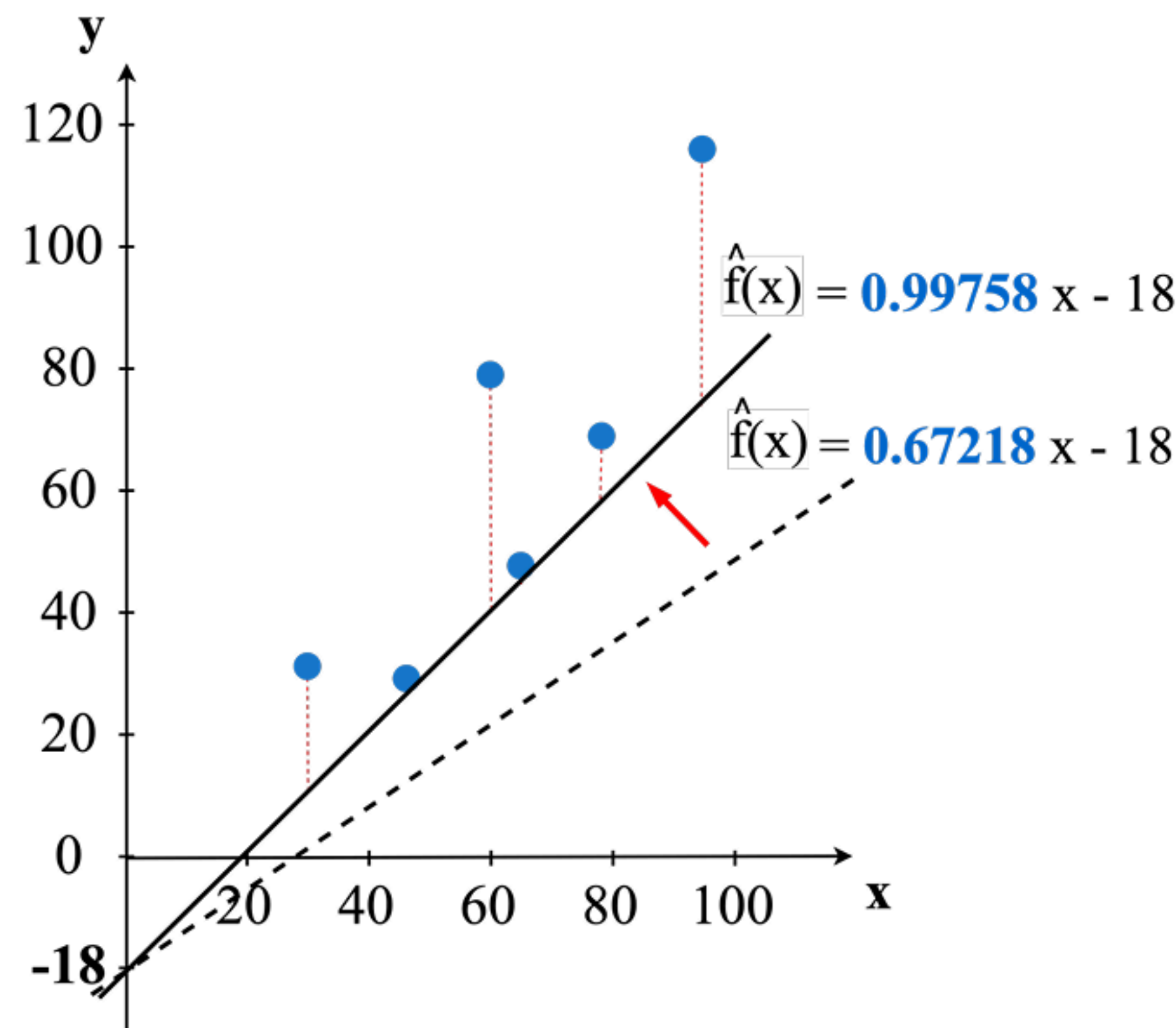


Descida do gradiente

Primeira iteração: $a_0 = 0$



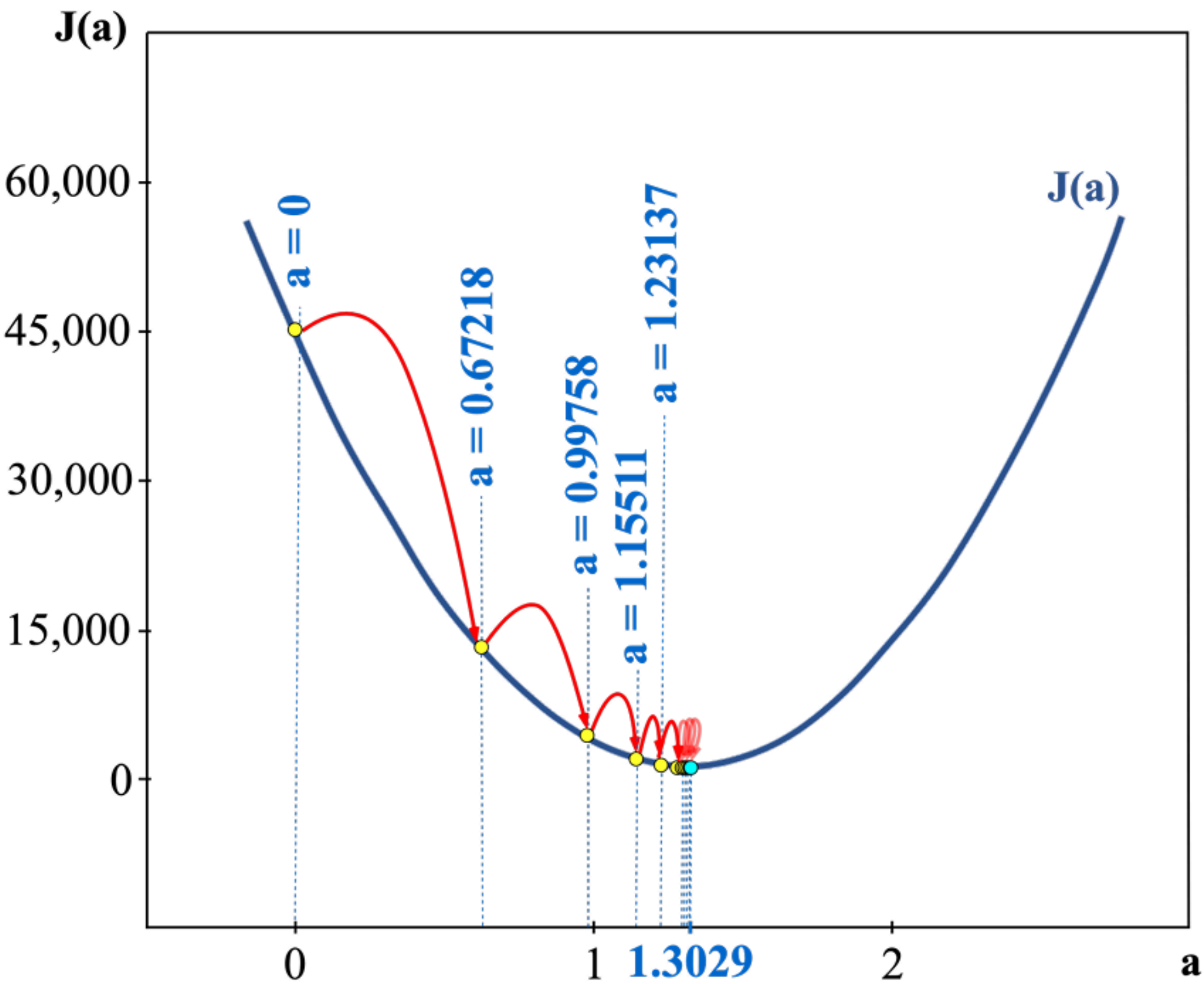
Segunda iteração: $a_1 = 0.67218$



Descida do gradiente

$$a_0 = 0$$

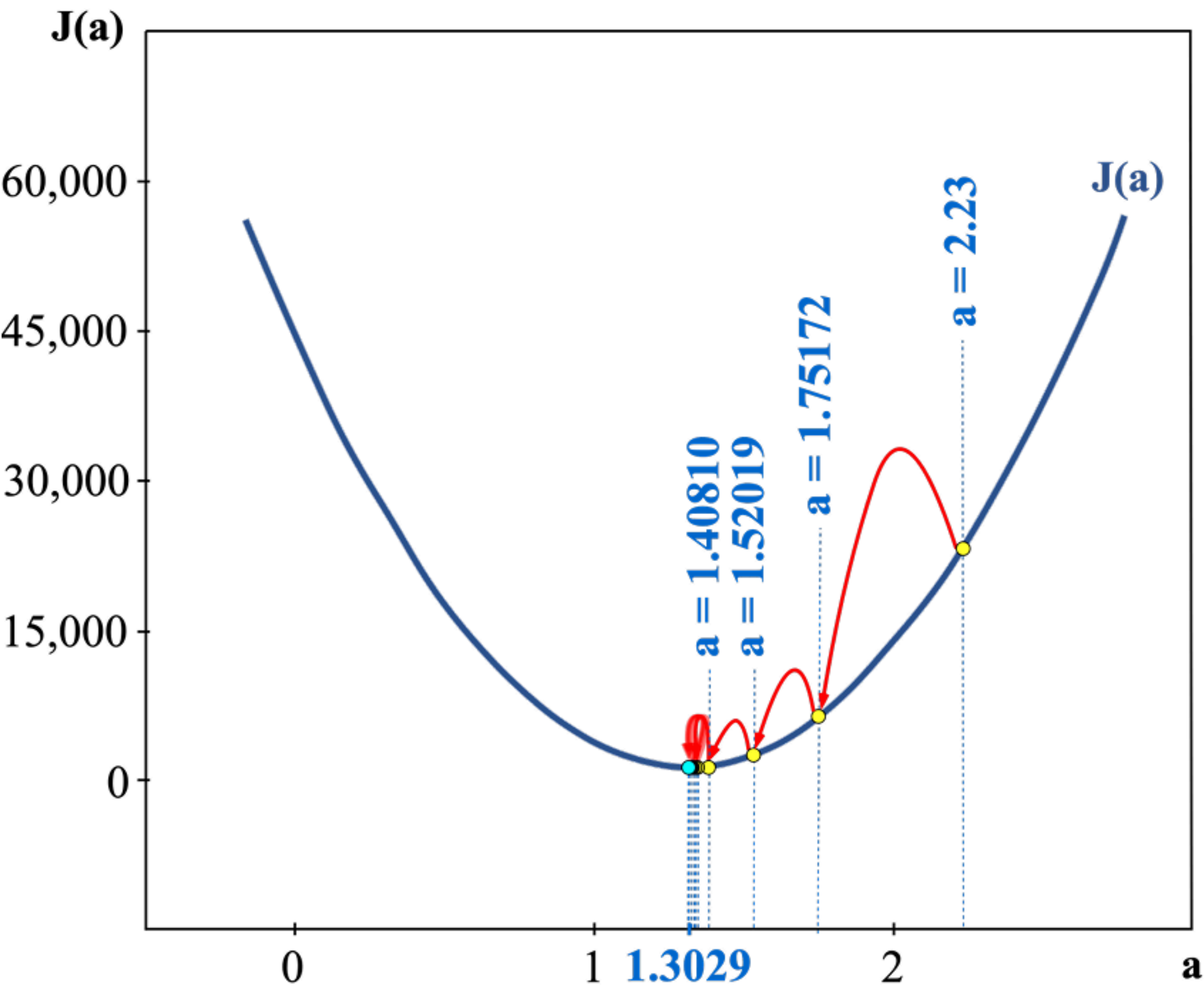
Iterations of Gradient Descent			
Iter	$\frac{\partial J}{\partial a}$	Step Size	a
1	-67218.0000	-0.67218	0.67218
2	-32540.2338	-0.32540	0.99758
3	-15752.7272	-0.15753	1.15511
4	-7625.8952	-0.07626	1.23137
5	-3691.6959	-0.03692	1.26829
6	-1787.1500	-0.01787	1.28616
7	-865.1593	-0.00865	1.29481
8	-418.8236	-0.00419	1.29900
9	-202.7525	-0.00203	1.30102
10	-98.1525	-0.00098	1.30201
11	-47.5156	-0.00048	1.30248
12	-23.0023	-0.00023	1.30271
13	-11.1354	-0.00011	1.30282
14	-5.3907	-0.00005	1.30288
15	-2.6096	-0.00003	1.30294



Descida do gradiente

$$a_0 = 2.23$$

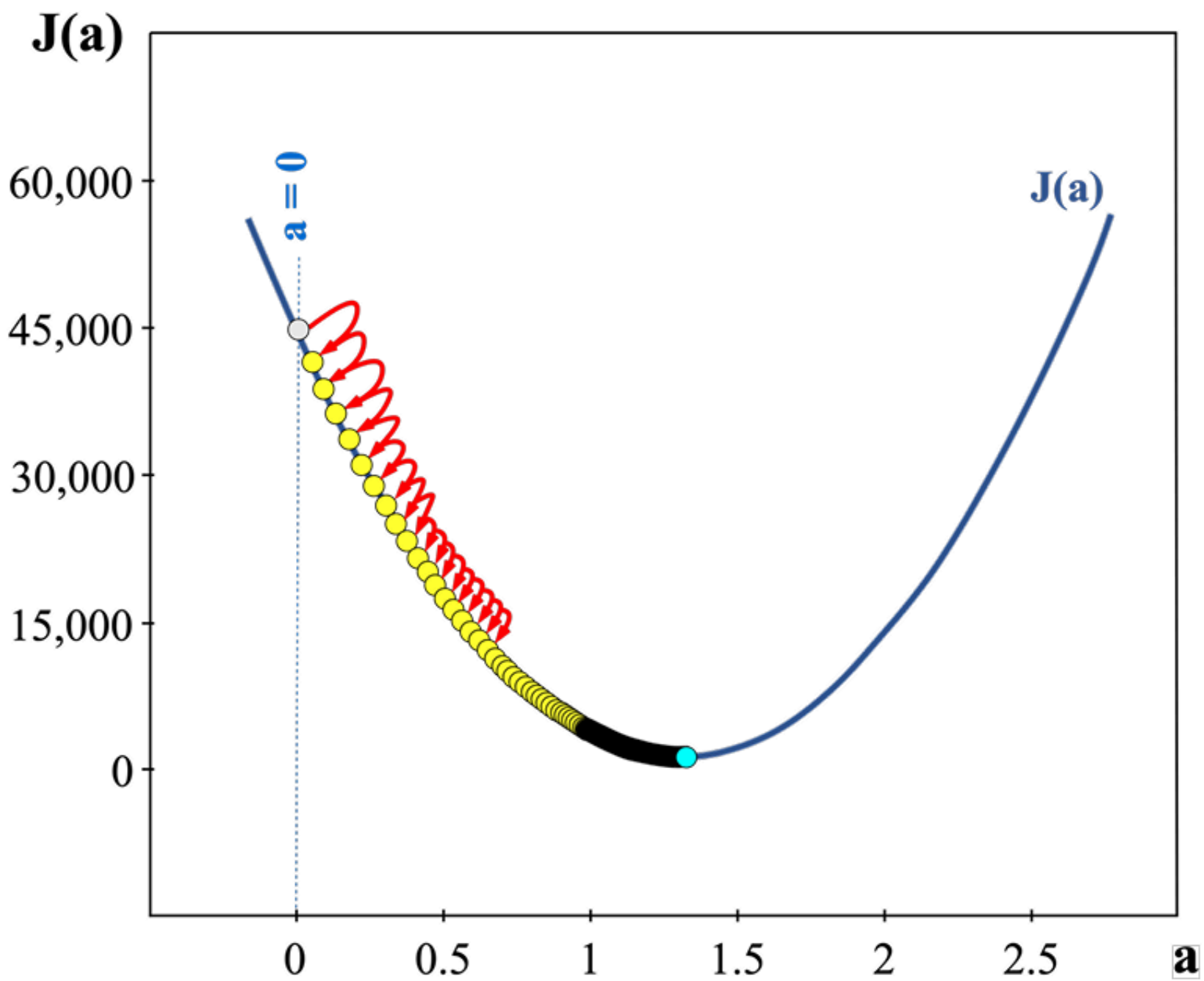
Gradient Descent Iterations			
Iter	$\frac{\partial J}{\partial a}$	Step Size	a
1	47827.7000	0.47828	1.75172
2	23153.3896	0.23153	1.52019
3	11208.5559	0.11209	1.40810
4	5426.0619	0.05426	1.35384
5	2626.7566	0.02627	1.32758
6	1271.6129	0.01272	1.31486
7	615.5878	0.00616	1.30870
8	298.006	0.00298	1.30572
9	144.2647	0.00144	1.30428
10	69.8386	0.00070	1.30358
11	33.8088	0.00034	1.30324
12	16.3669	0.00016	1.30308
13	7.9232	0.00008	1.30300
14	3.8356	0.00004	1.30296
15	1.8568	0.00002	1.30294



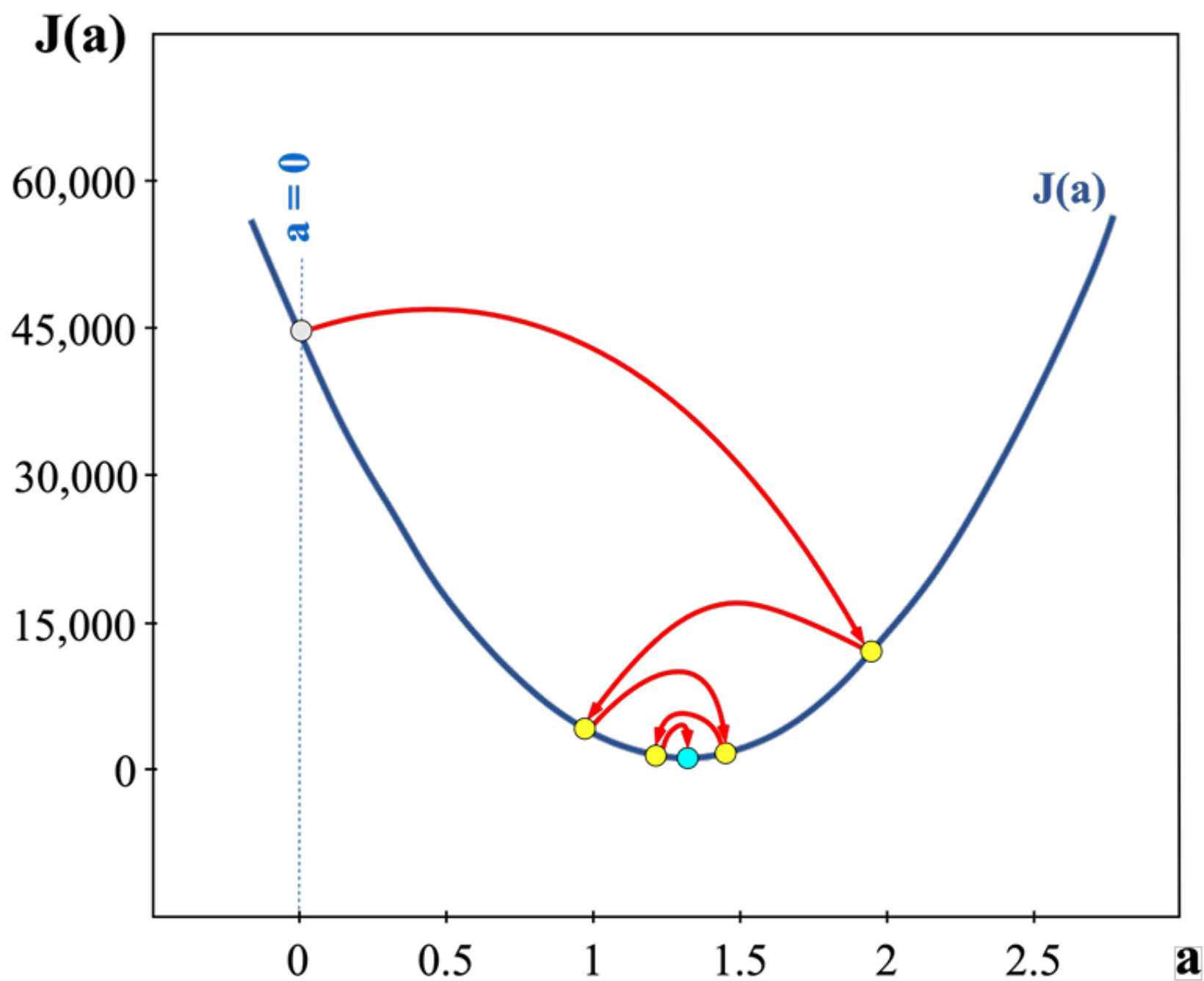
Descida do gradiente

$$a_{new} = a - \underbrace{\frac{\partial J}{\partial a}}_{\text{Step Size}} \cdot l$$

Taxa de aprendizado baixa

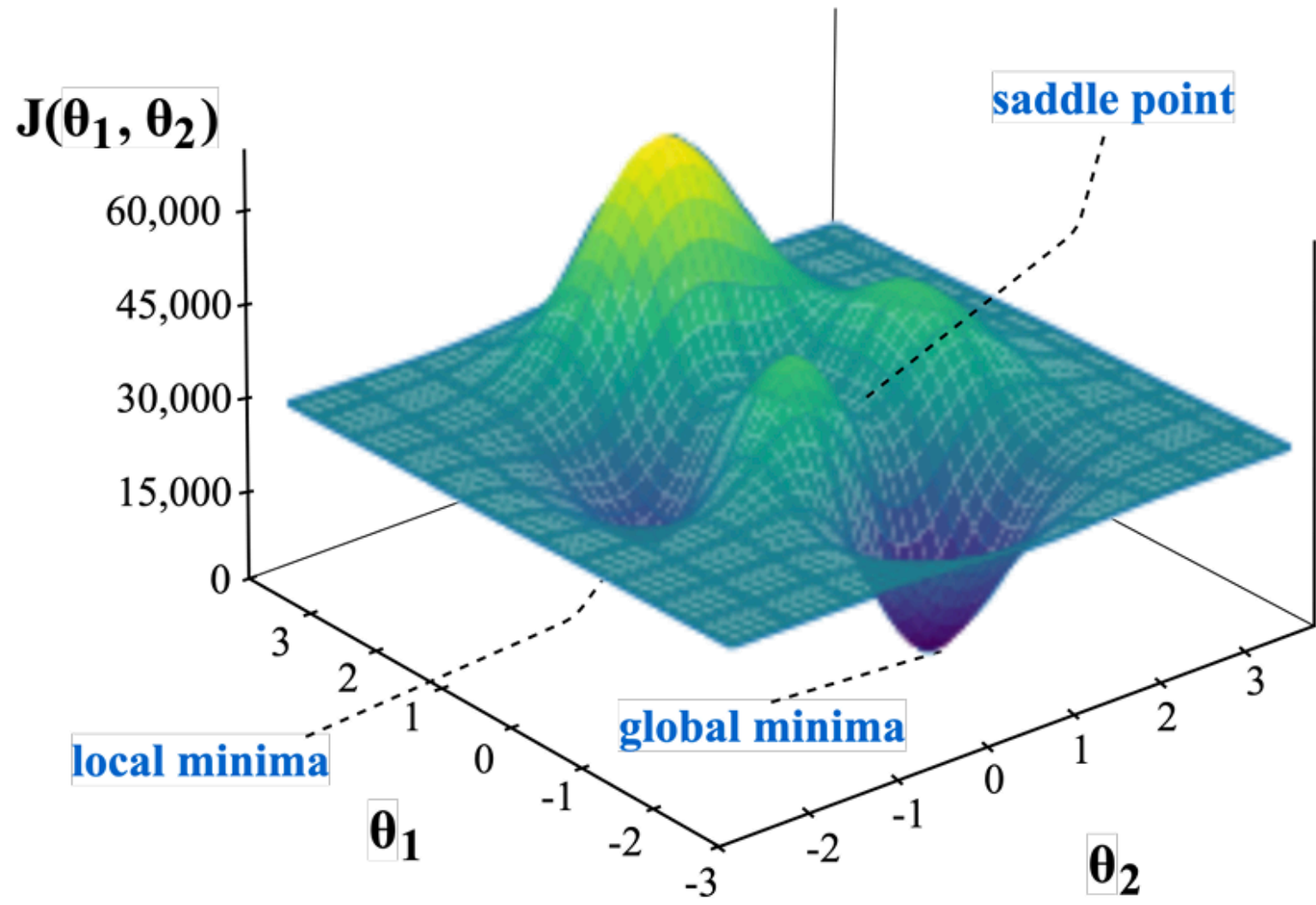
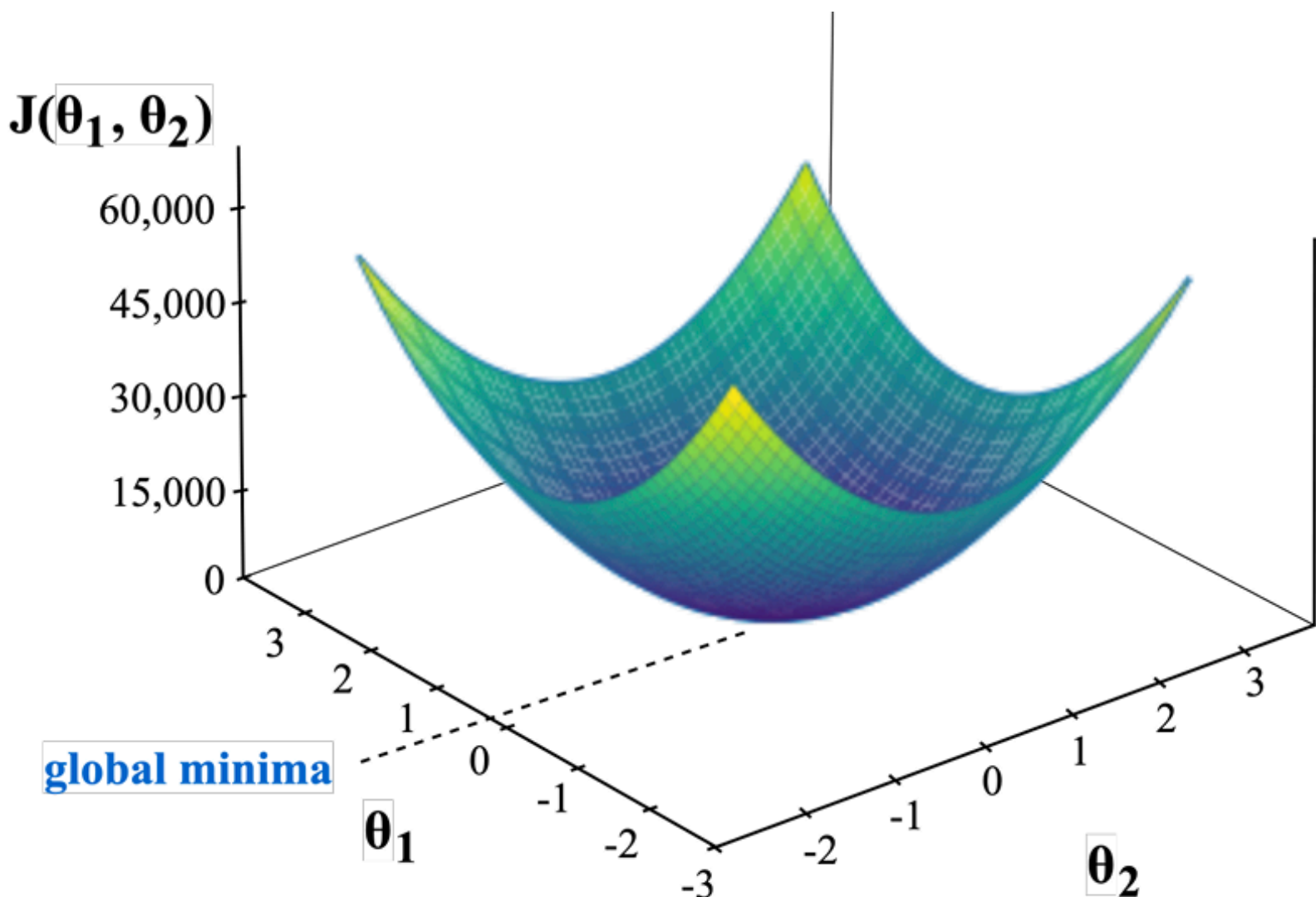
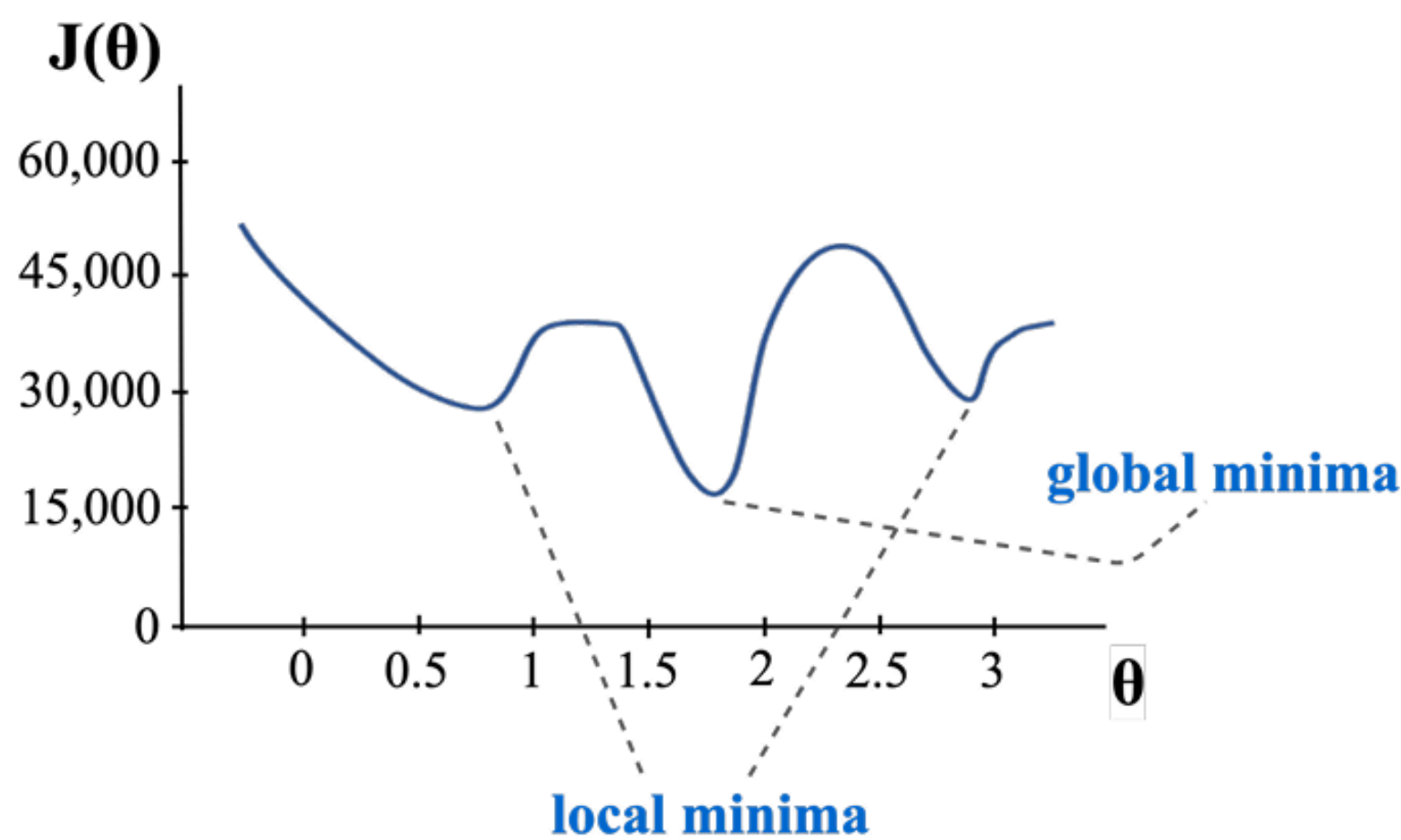
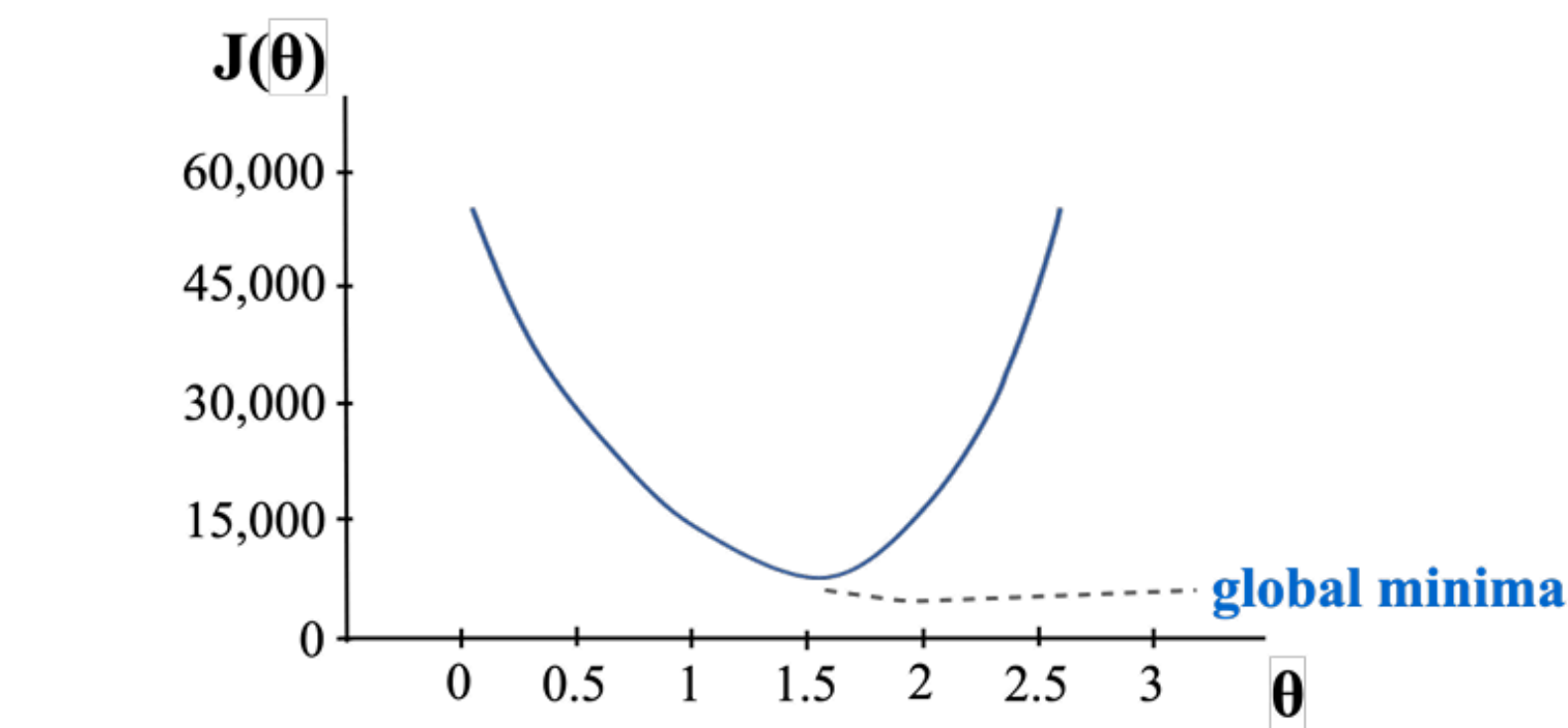


Taxa de aprendizado alta



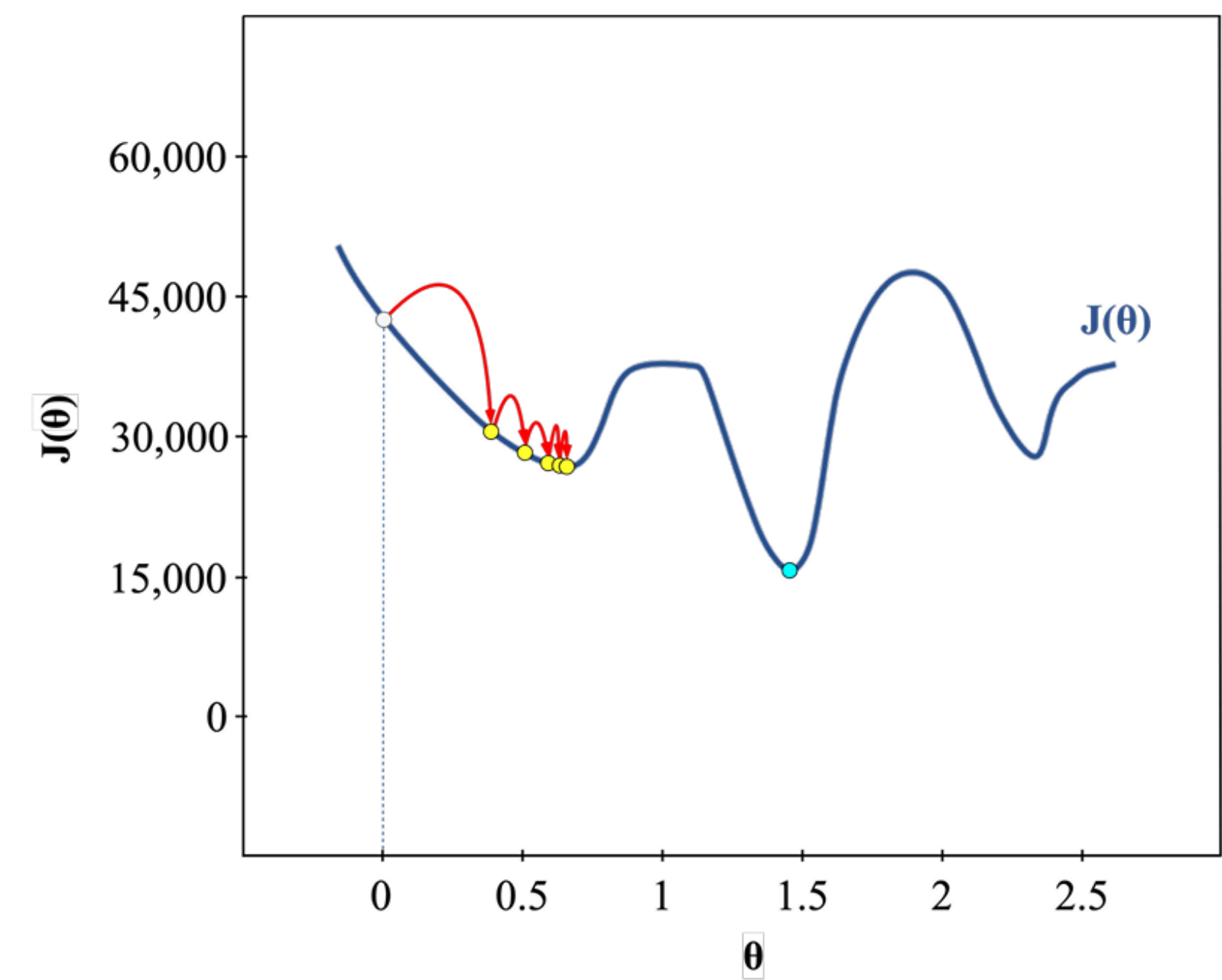
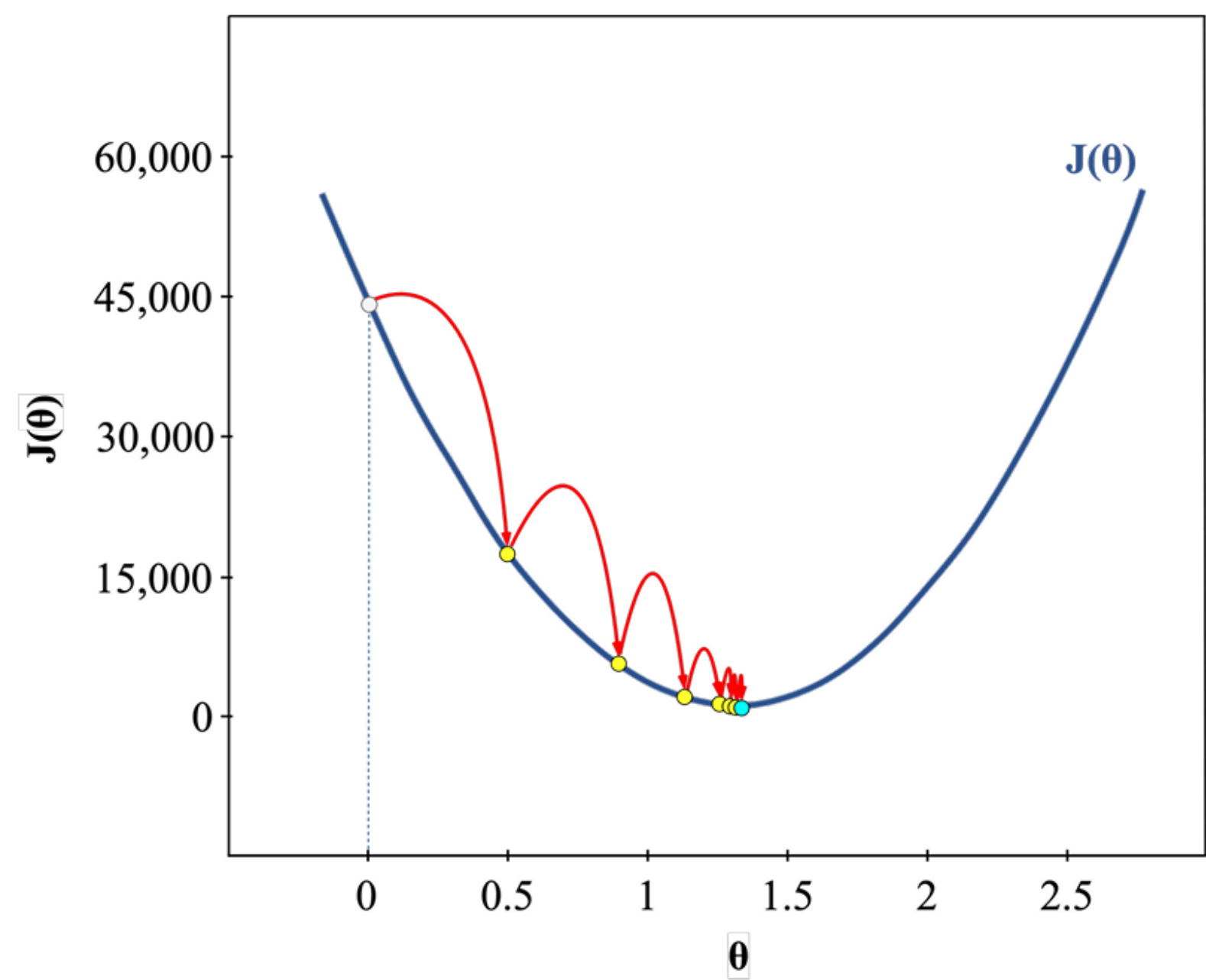
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Mínimos locais e globais

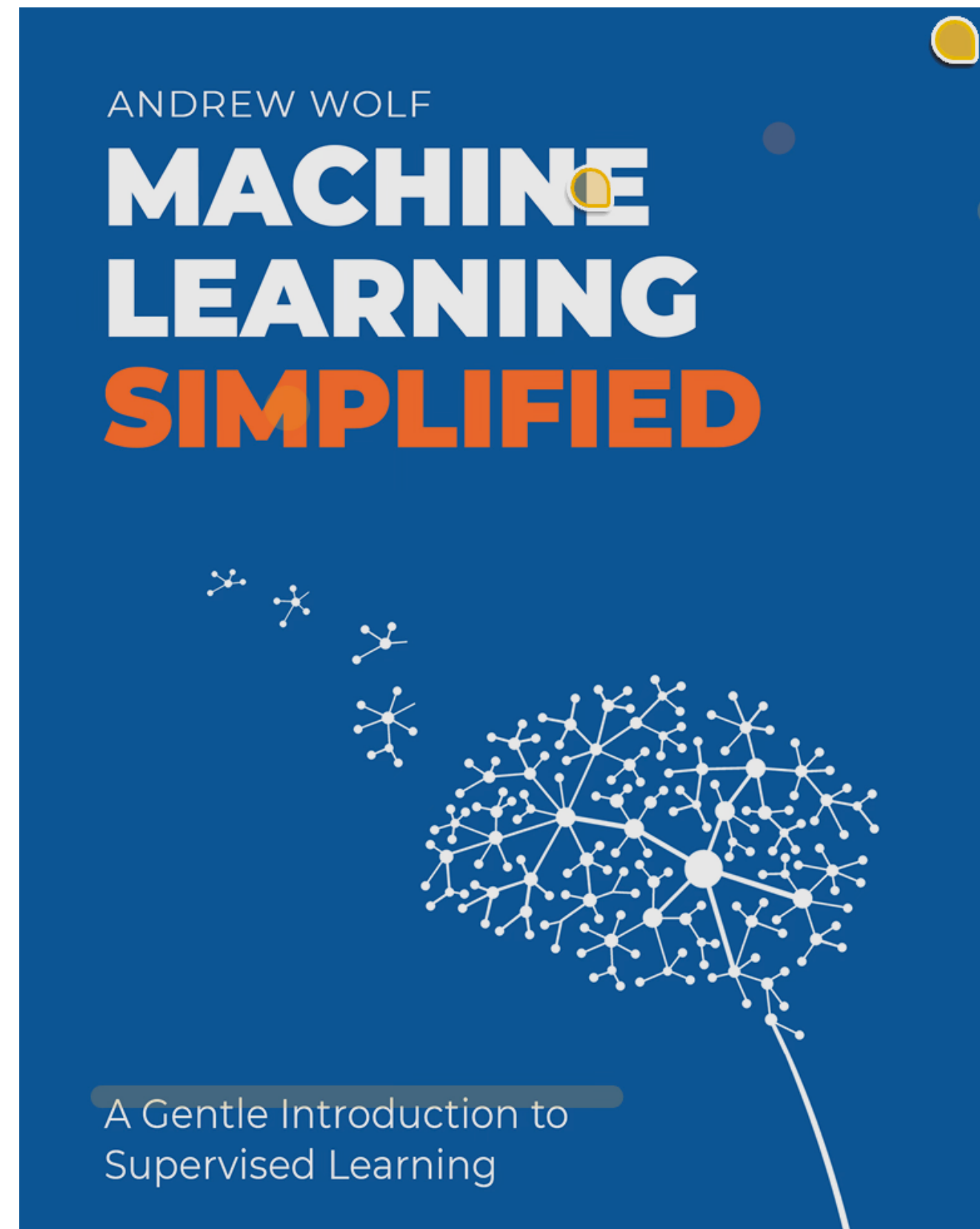


Descida do gradiente

Mínimos locais e globais



Leitura adicional



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